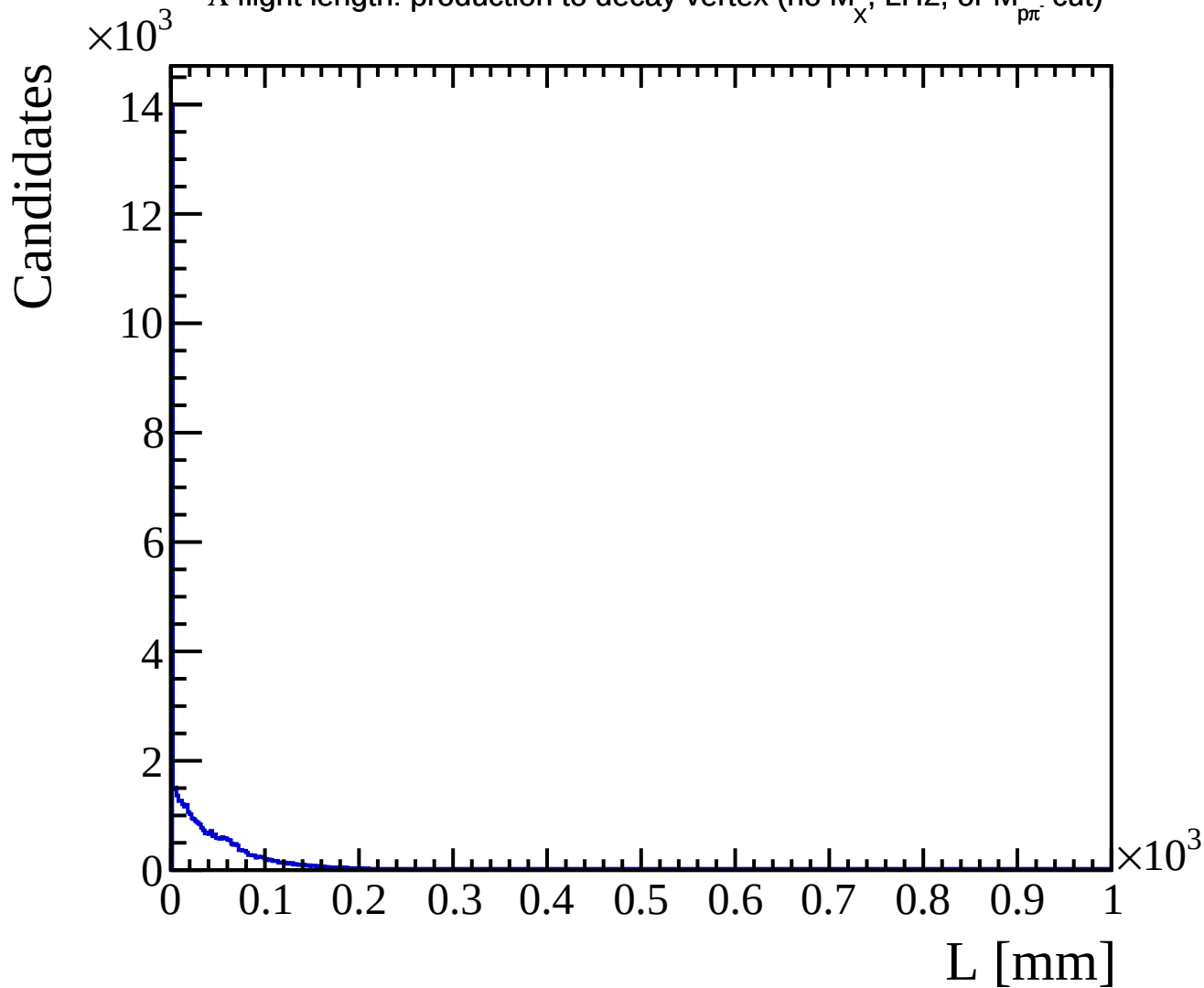
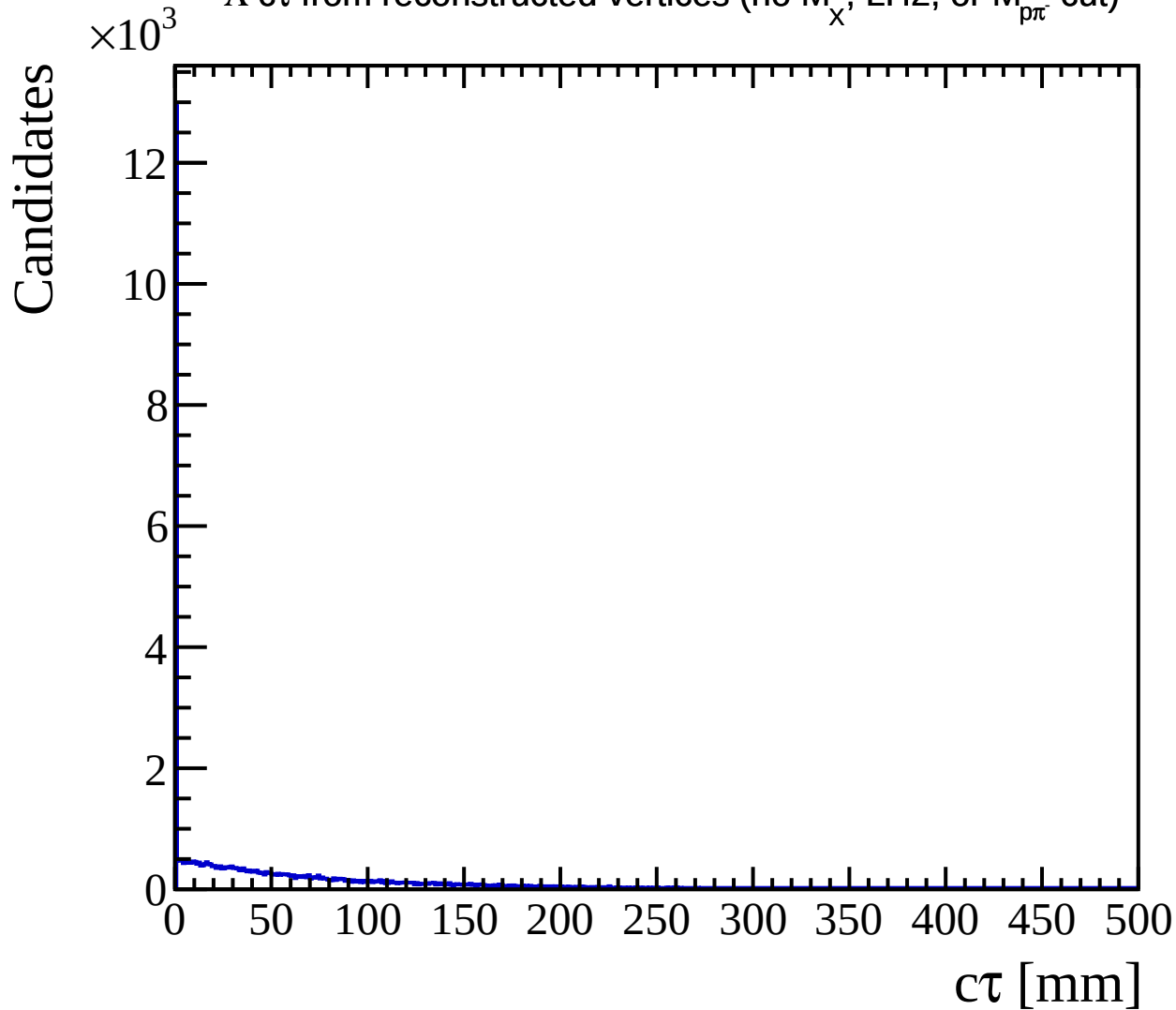


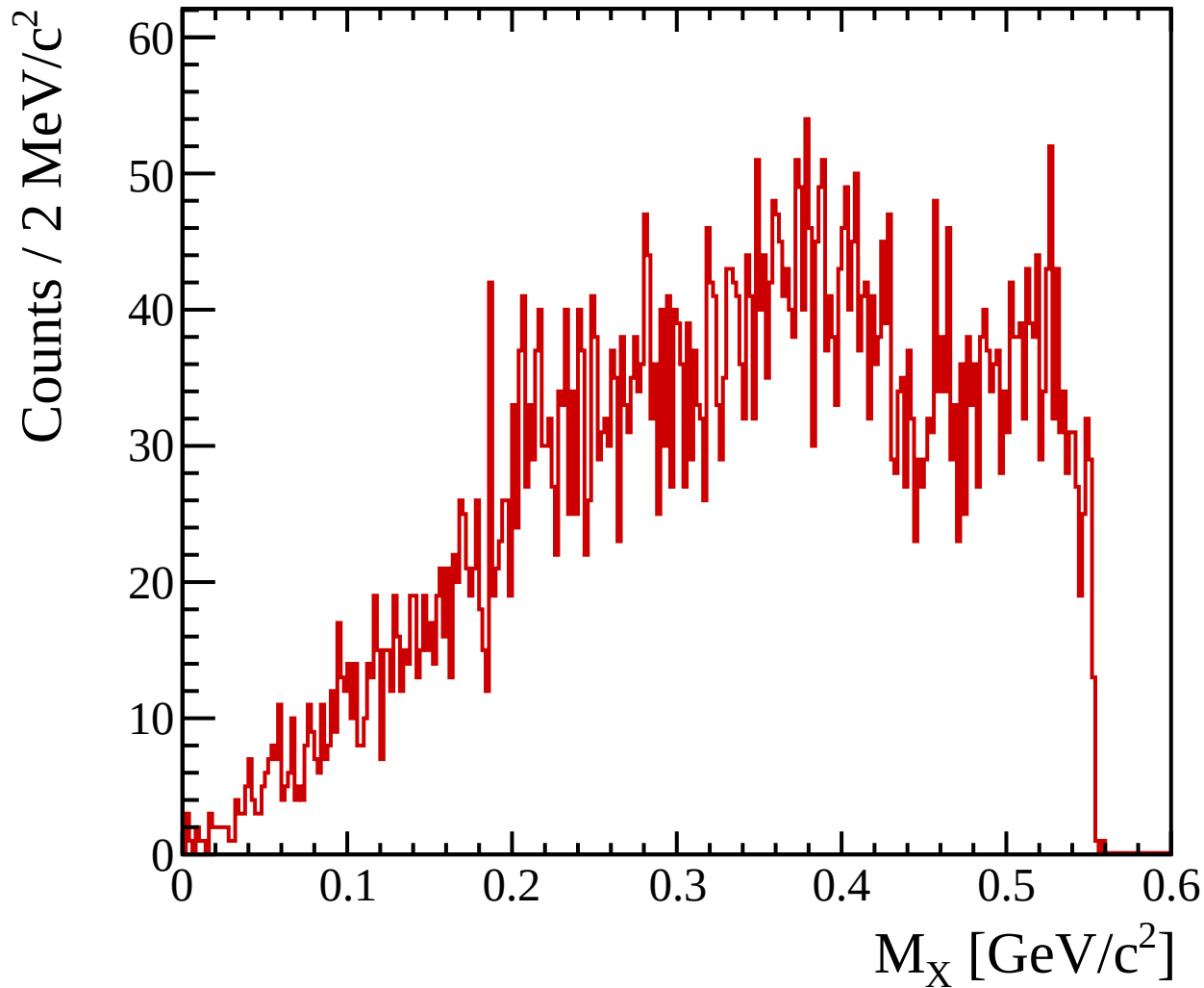
Λ flight length: production to decay vertex (no M_X , LH2, or $M_{\rho\pi^-}$ cut)



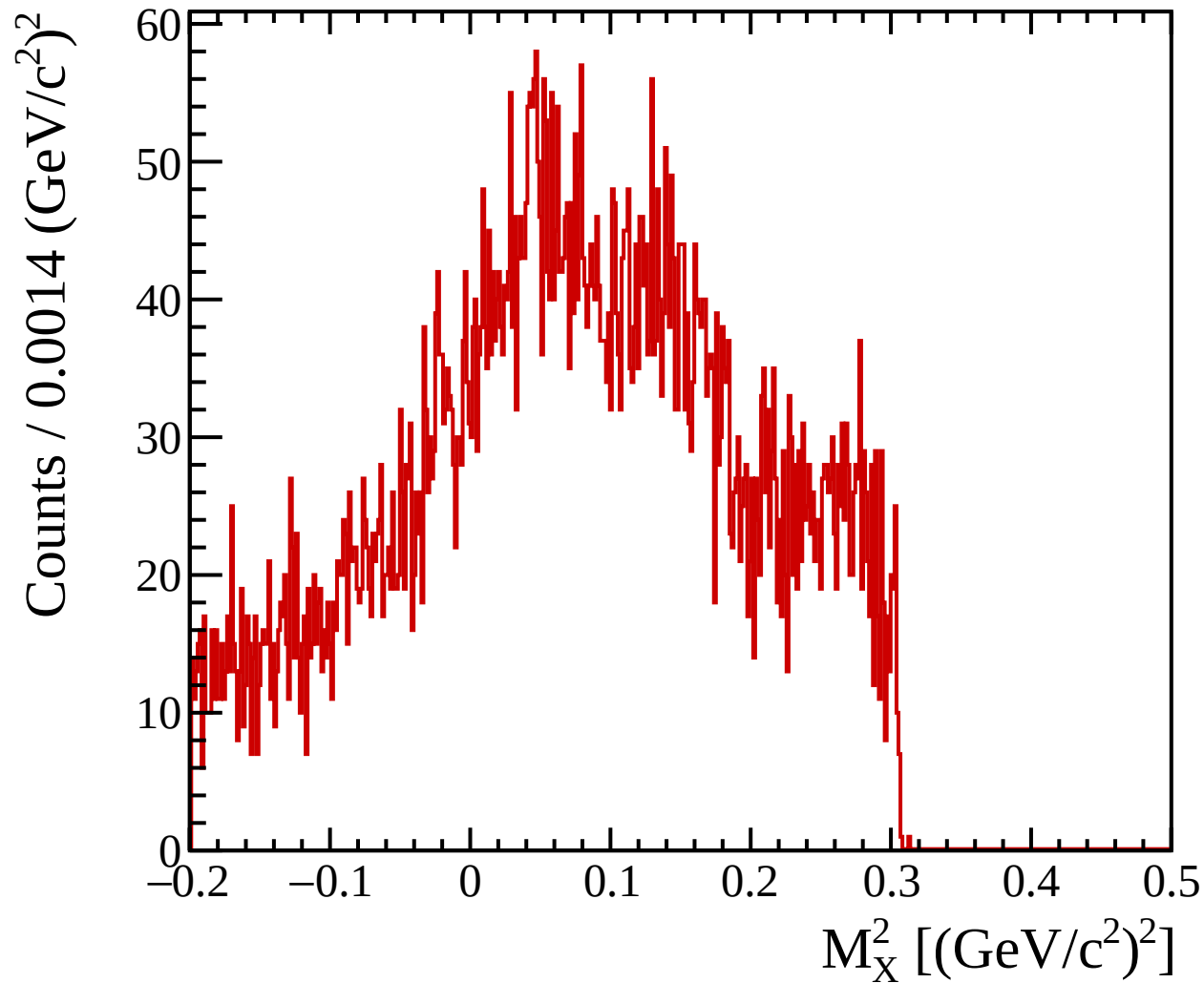
Λ $c\tau$ from reconstructed vertices (no M_{χ^-} , LH2, or $M_{\rho\pi^-}$ cut)

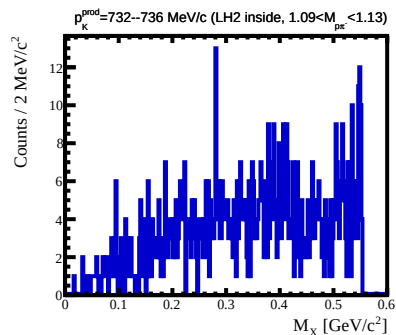
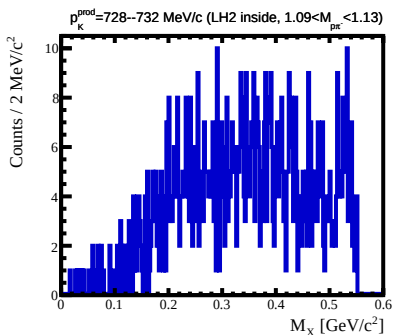
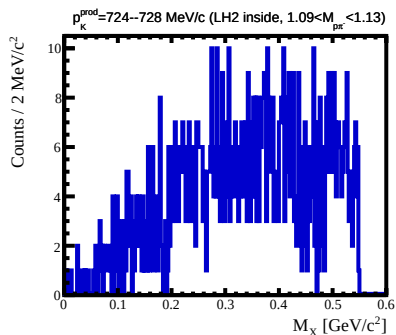
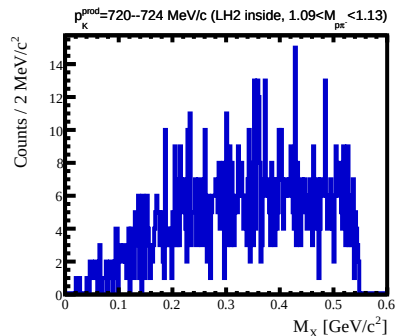
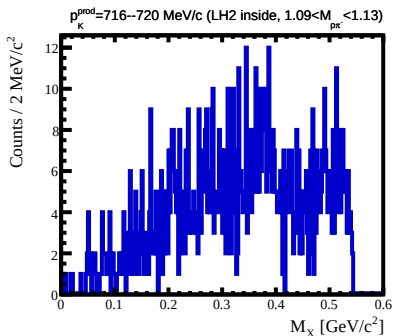
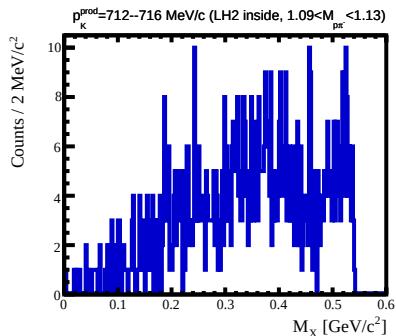
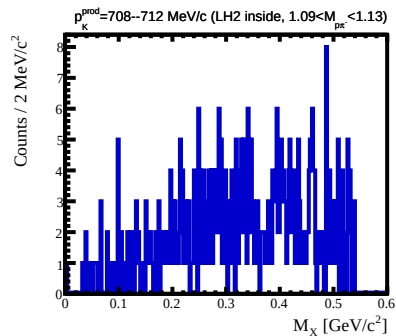
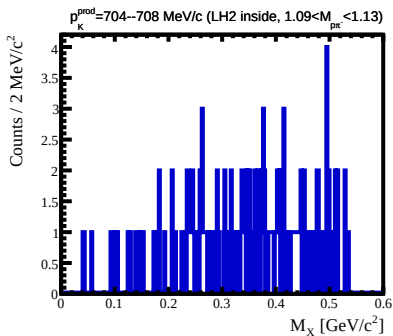
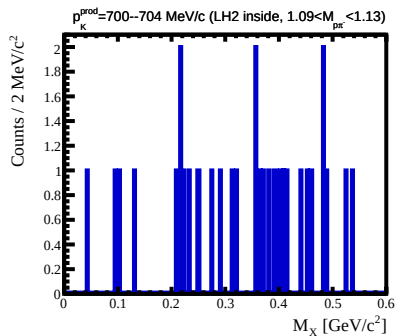


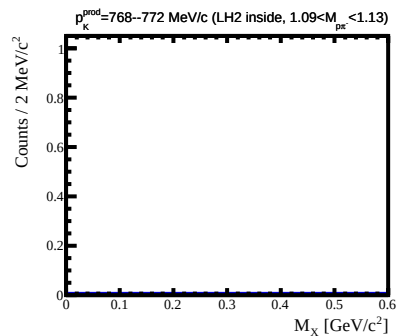
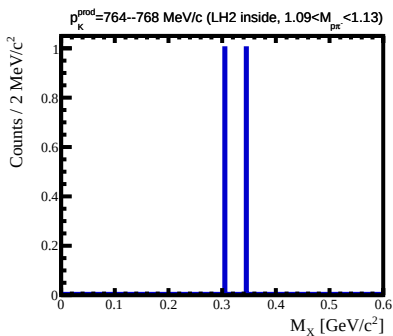
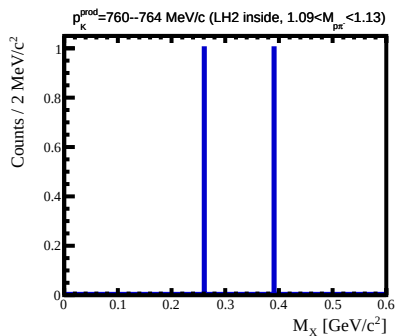
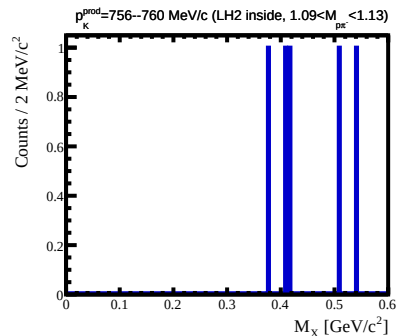
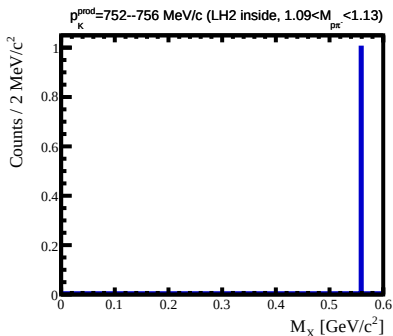
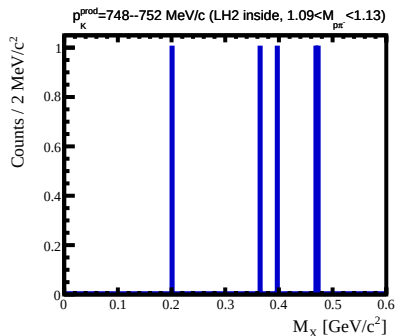
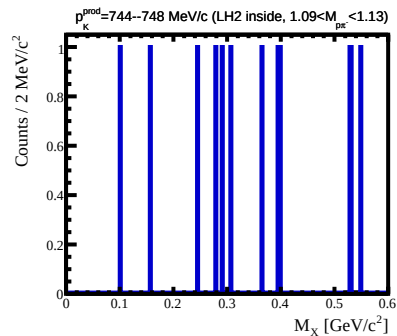
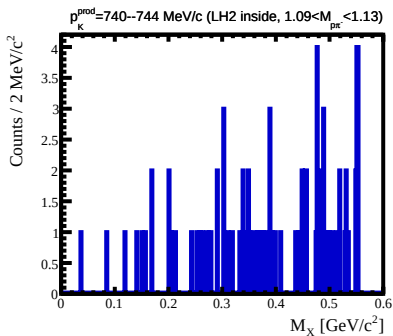
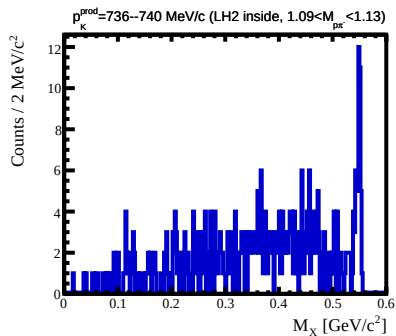
Λ missing mass: LH2 inside, $1.09 < M_{p\pi^-} < 1.13$

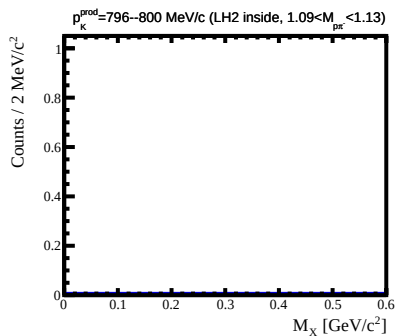
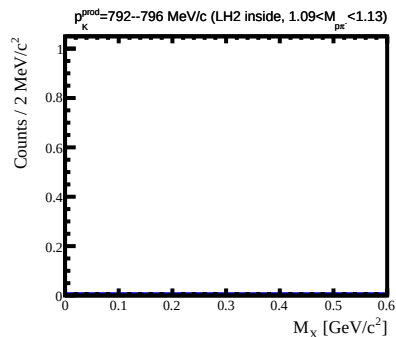
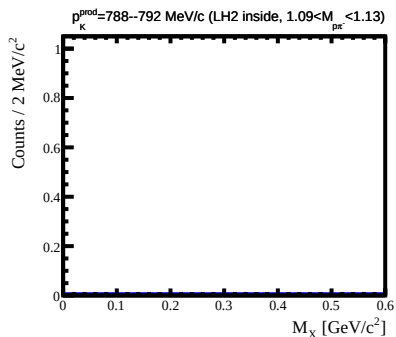
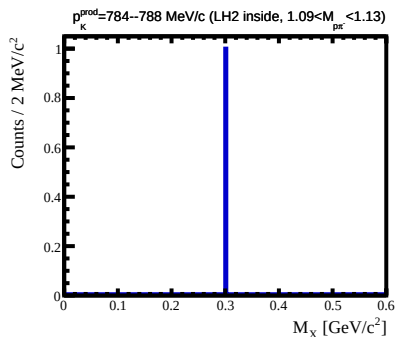
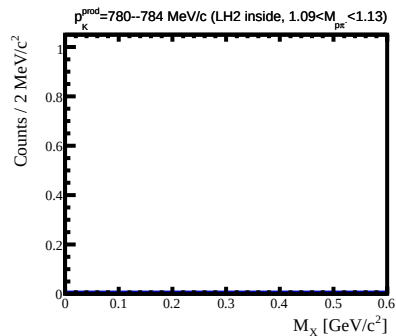
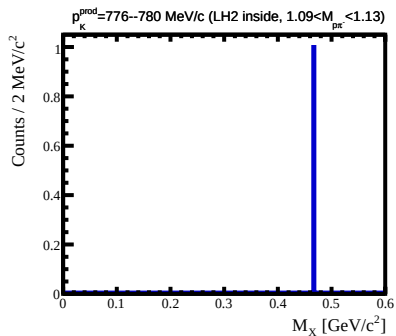
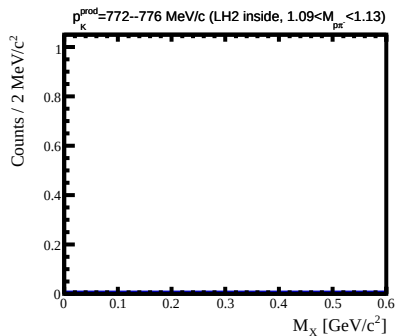


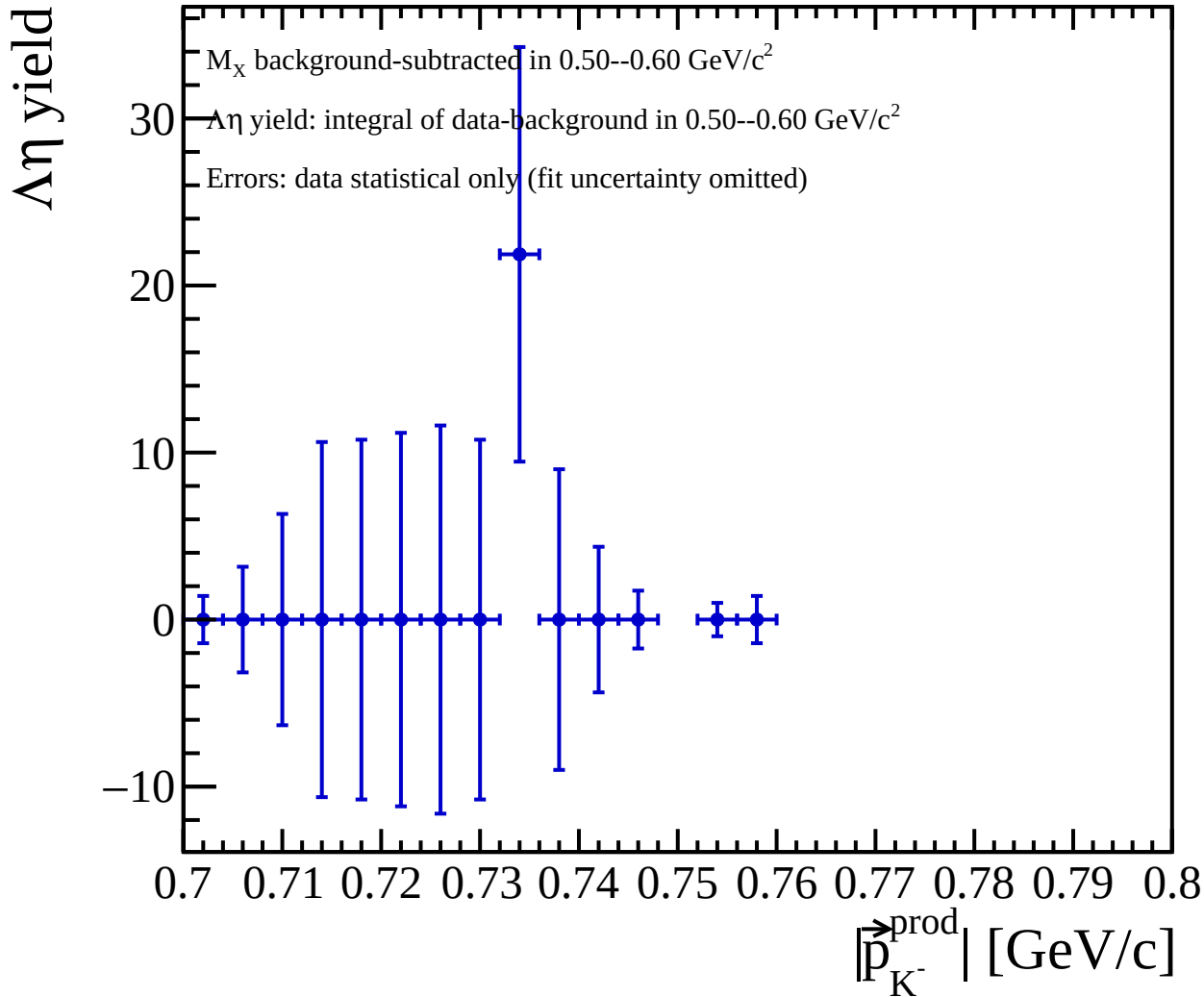
Λ missing mass squared: LH2 inside, $1.09 < M_{p\pi^-} < 1.13$



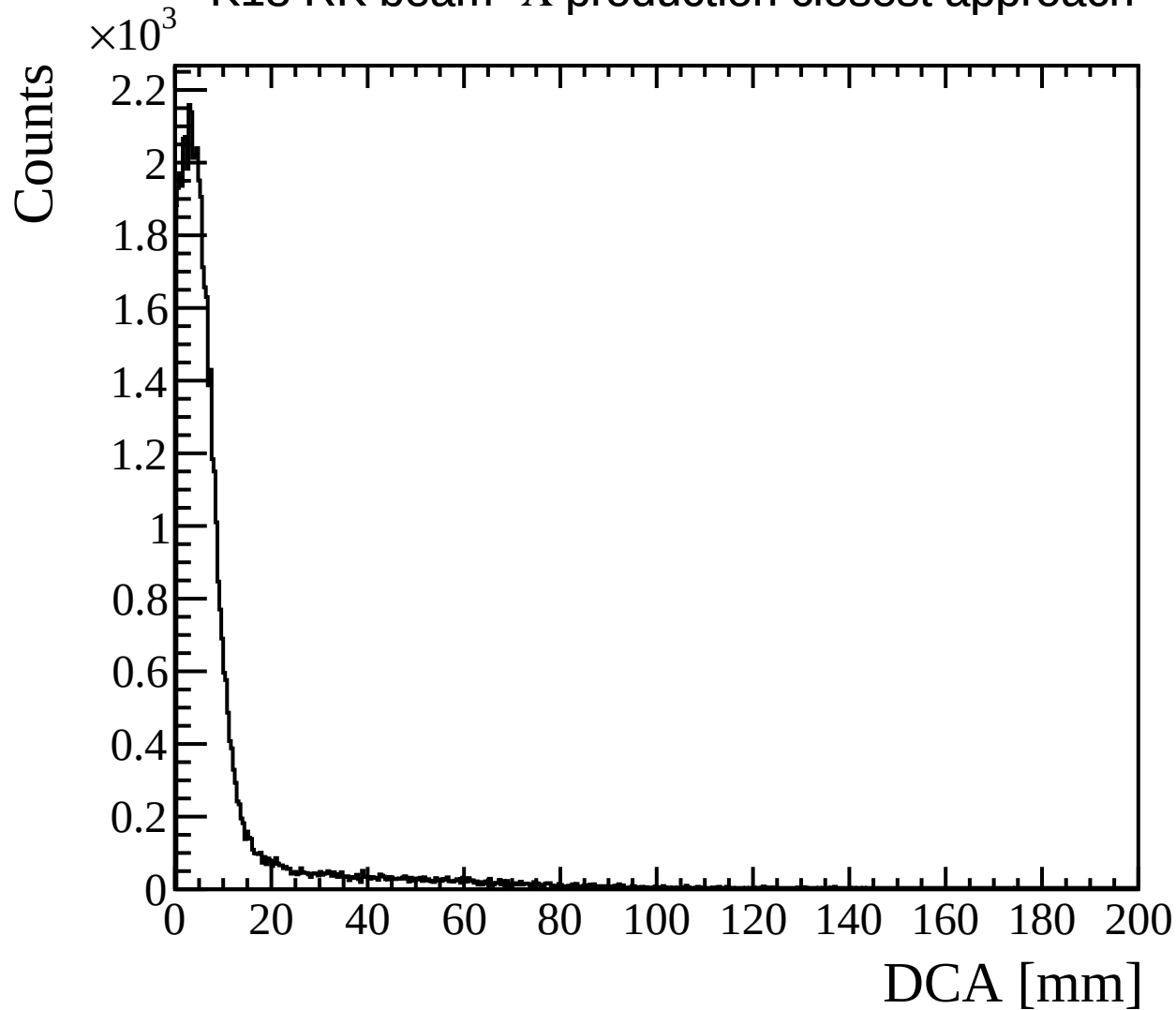




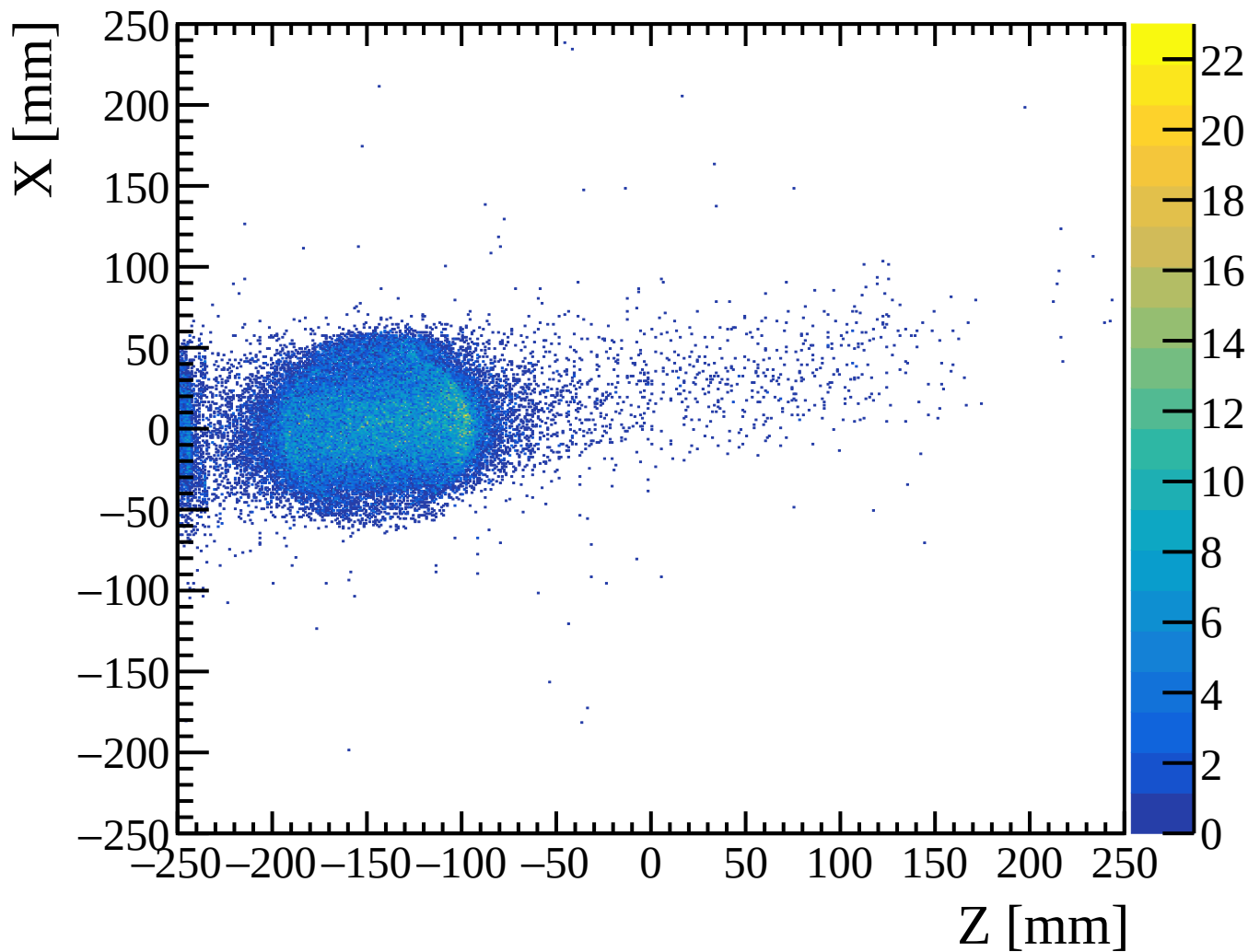




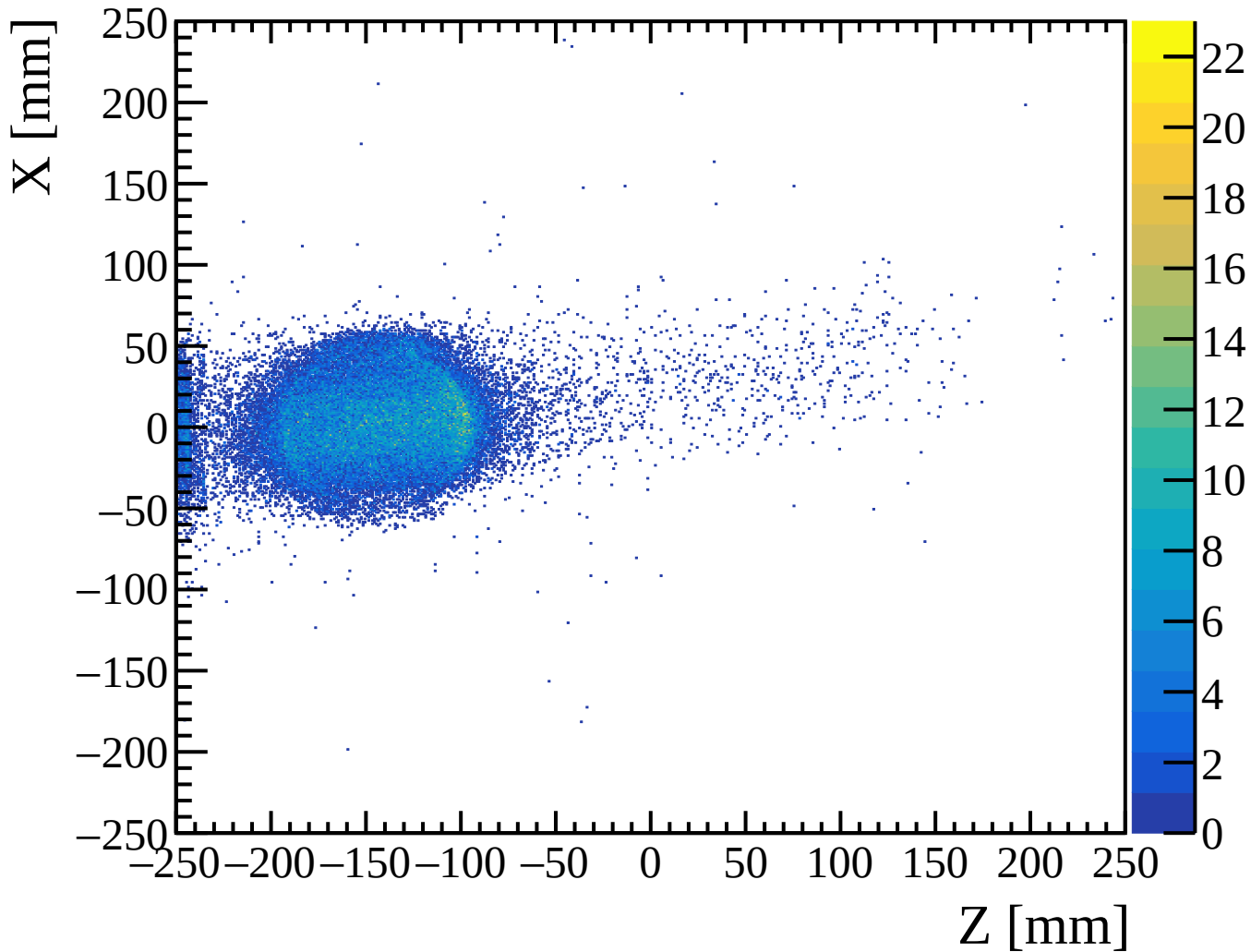
K18 RK beam-- Λ production closest approach



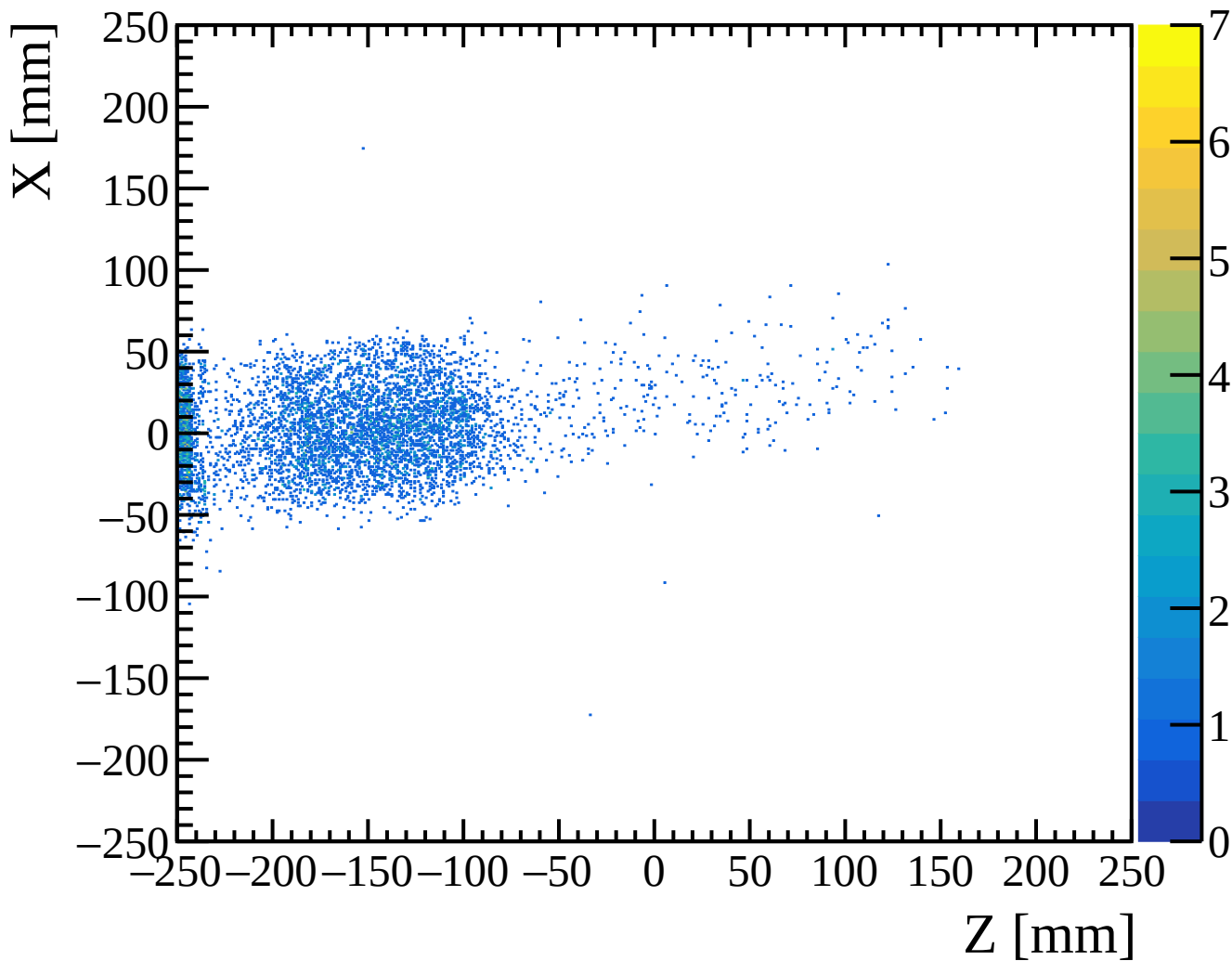
Reconstructed Λ production vertex (DST inclusive)



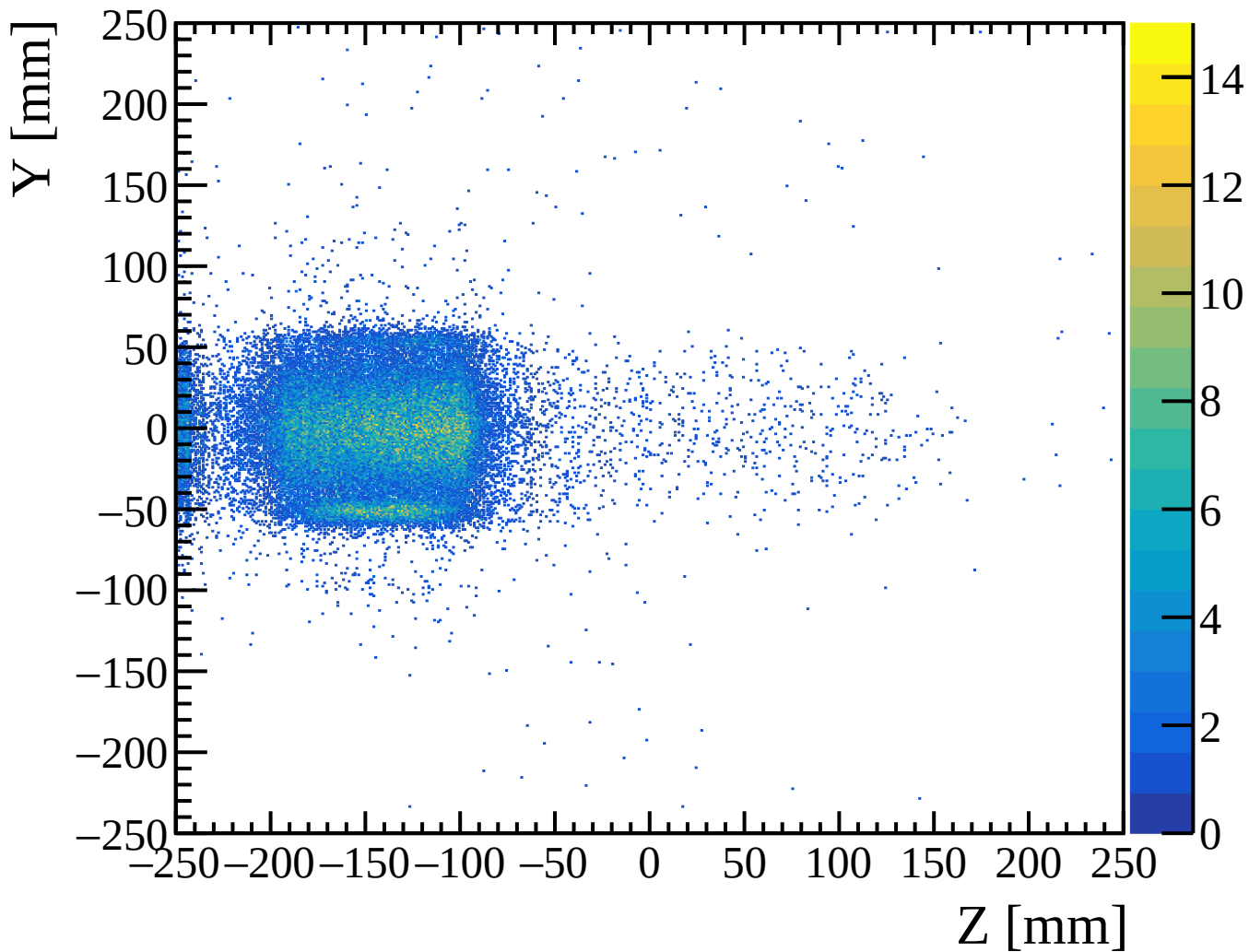
Reconstructed Λ production vertex (no LH2/Lambda-mass/ M_x cut)



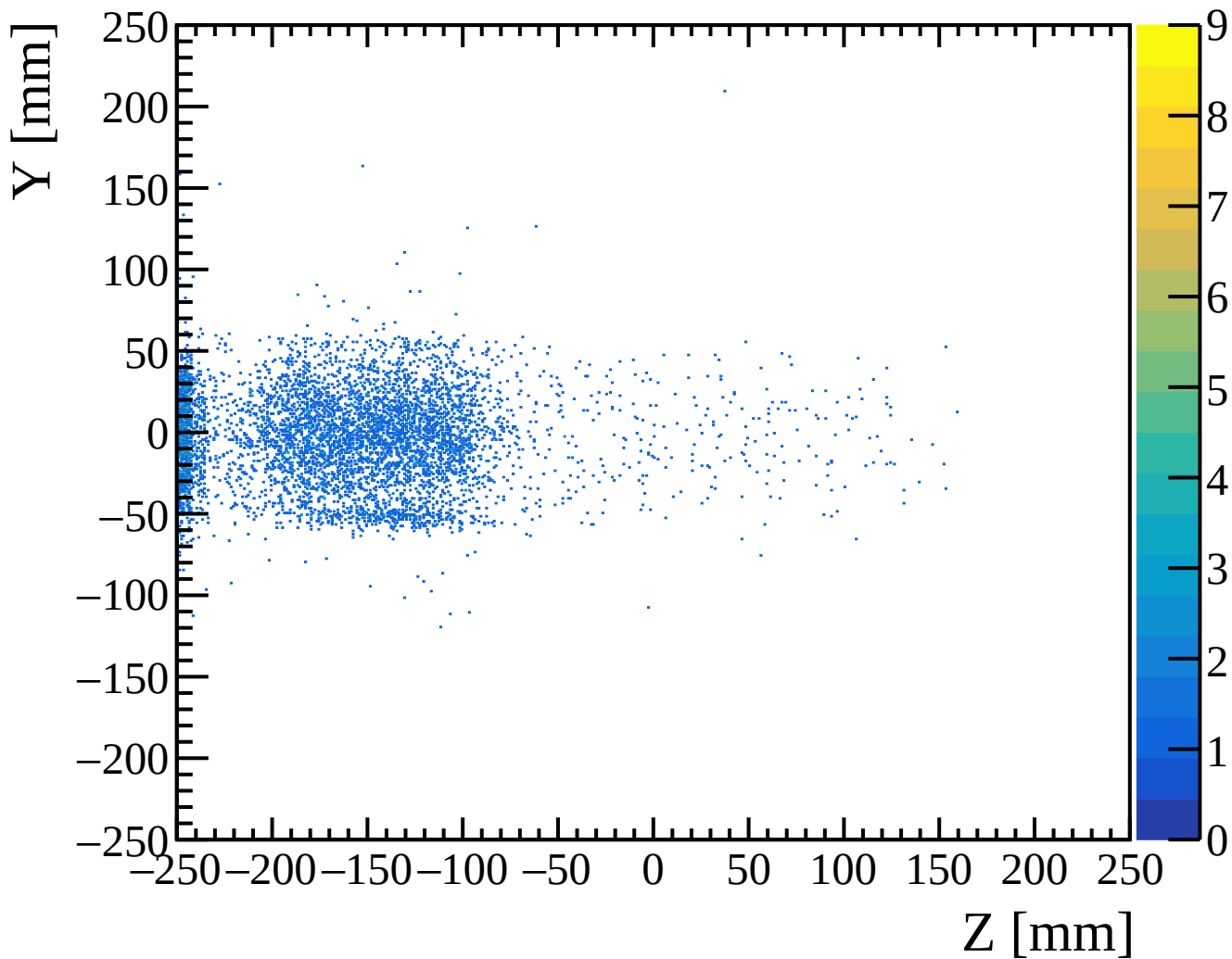
Reconstructed Λ production vertex (no LH2/Lambda-mass cut, $M_x > 0.50$)



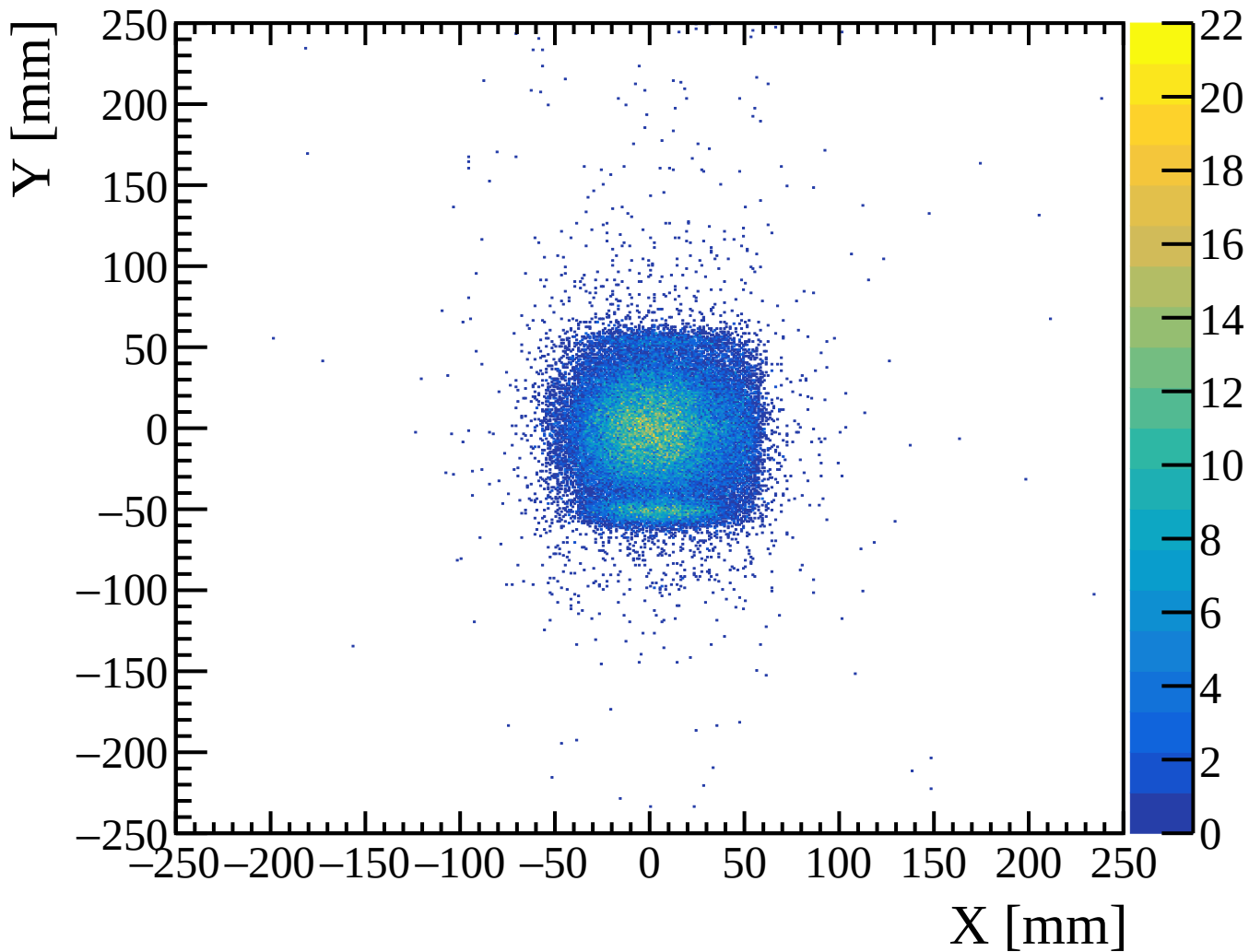
Reconstructed Λ production vertex (no LH2/Lambda-mass/ M_x cut)



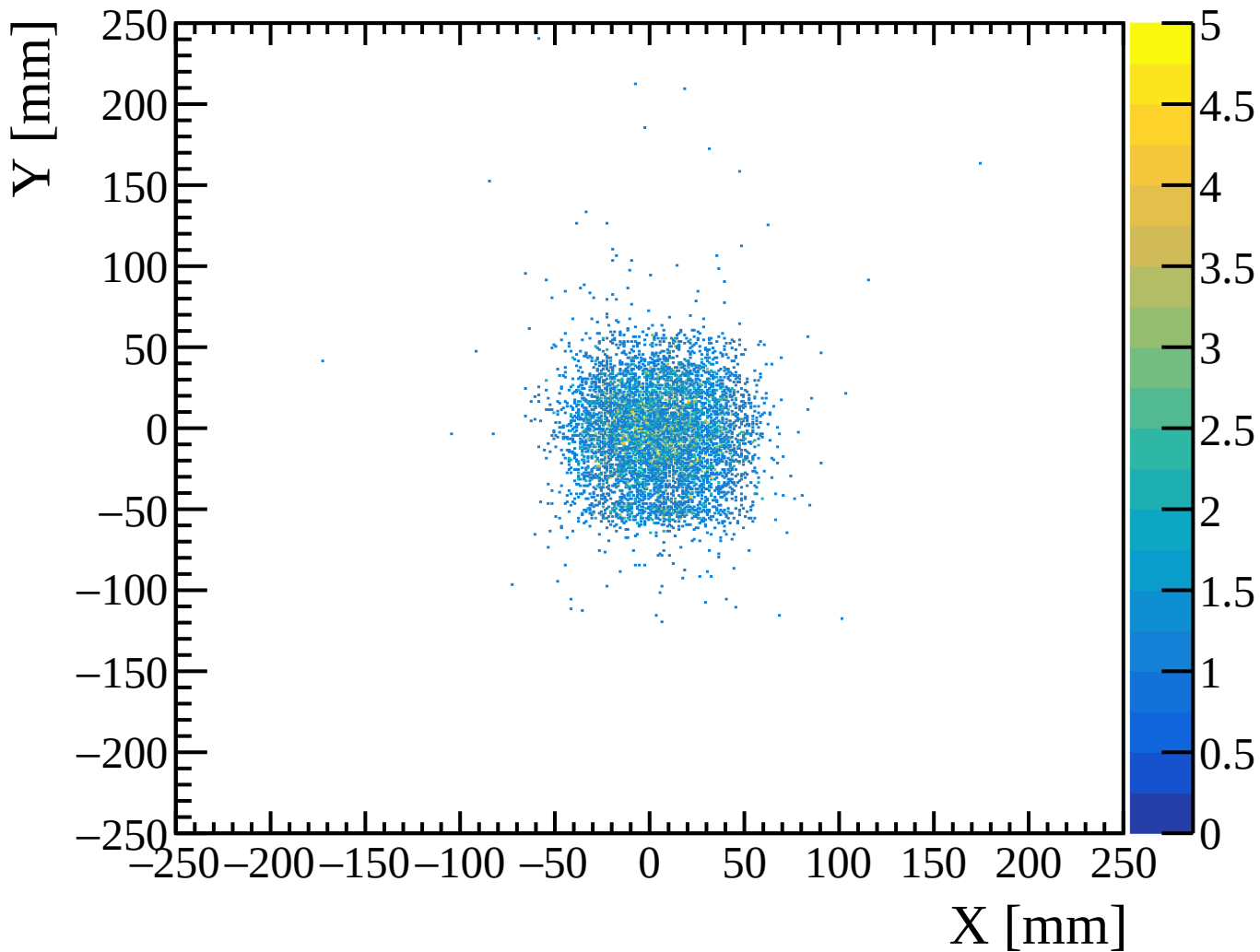
Reconstructed Λ production vertex (no LH2/Lambda-mass cut, $M_x > 0.50$)

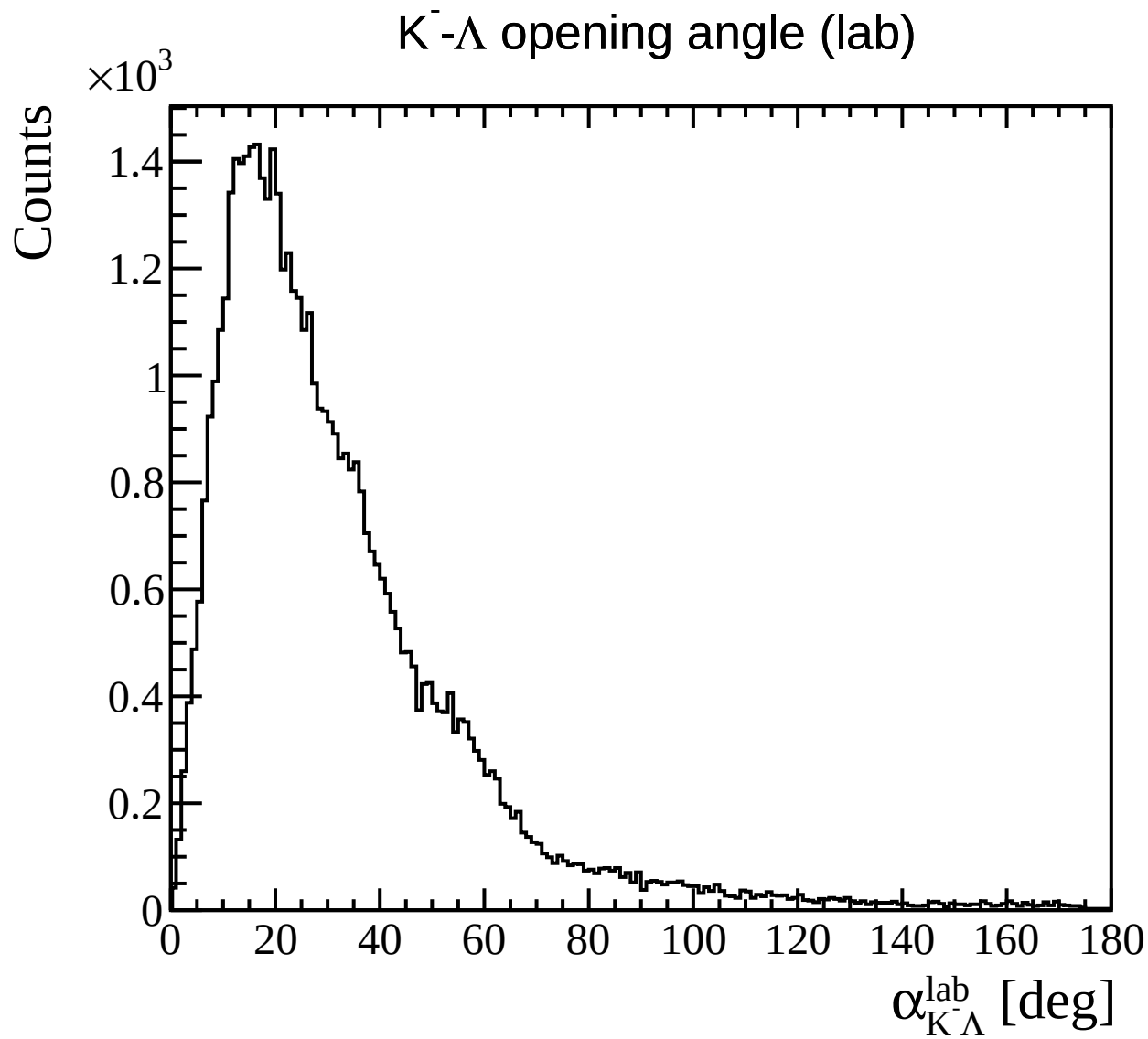


Reconstructed Λ production vertex (no LH2/Lambda-mass/ M_X cut)

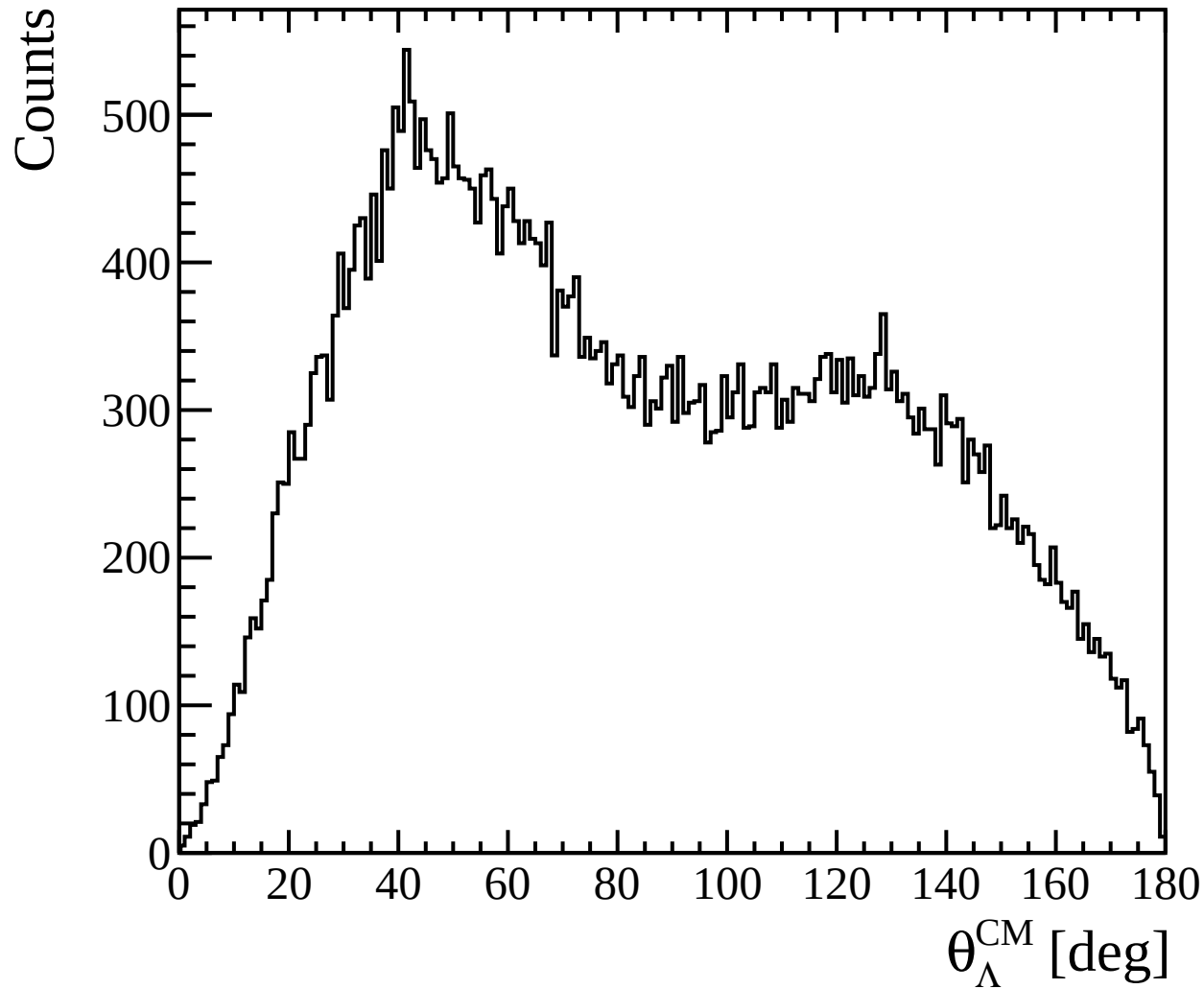


Reconstructed Λ production vertex (no LH2/Lambda-mass cut, $M_x > 0.50$)

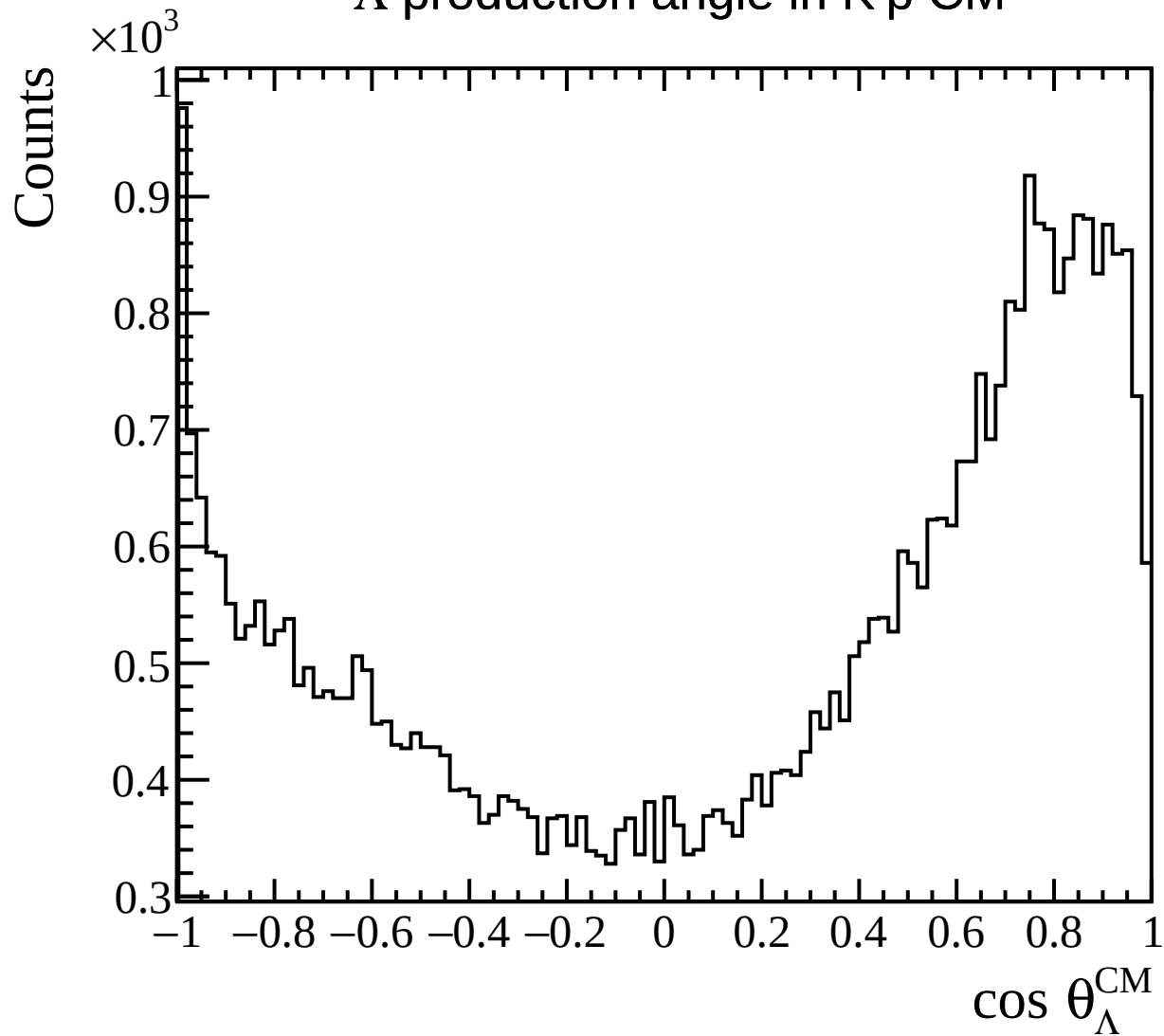




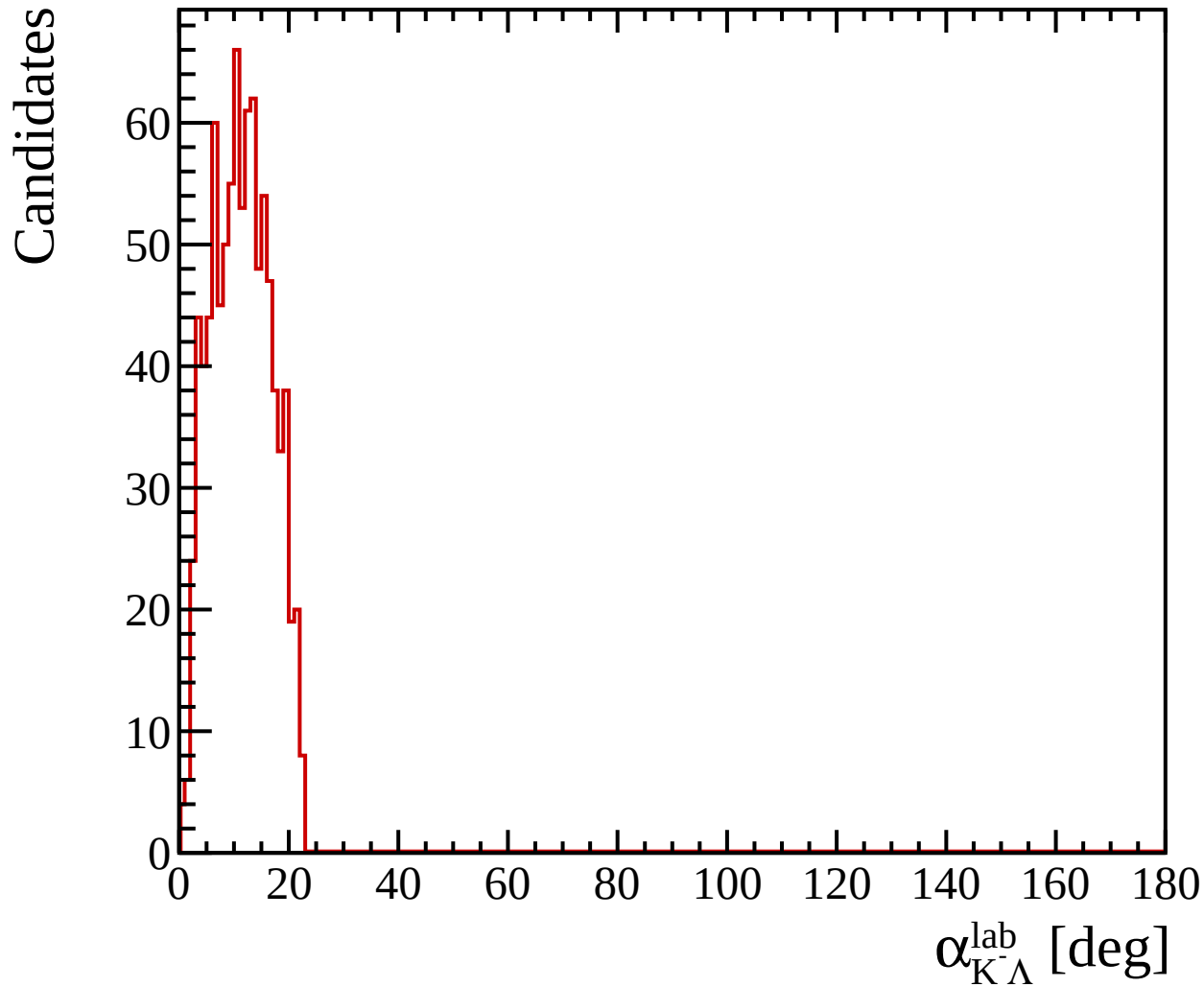
Λ production angle in K^-p CM



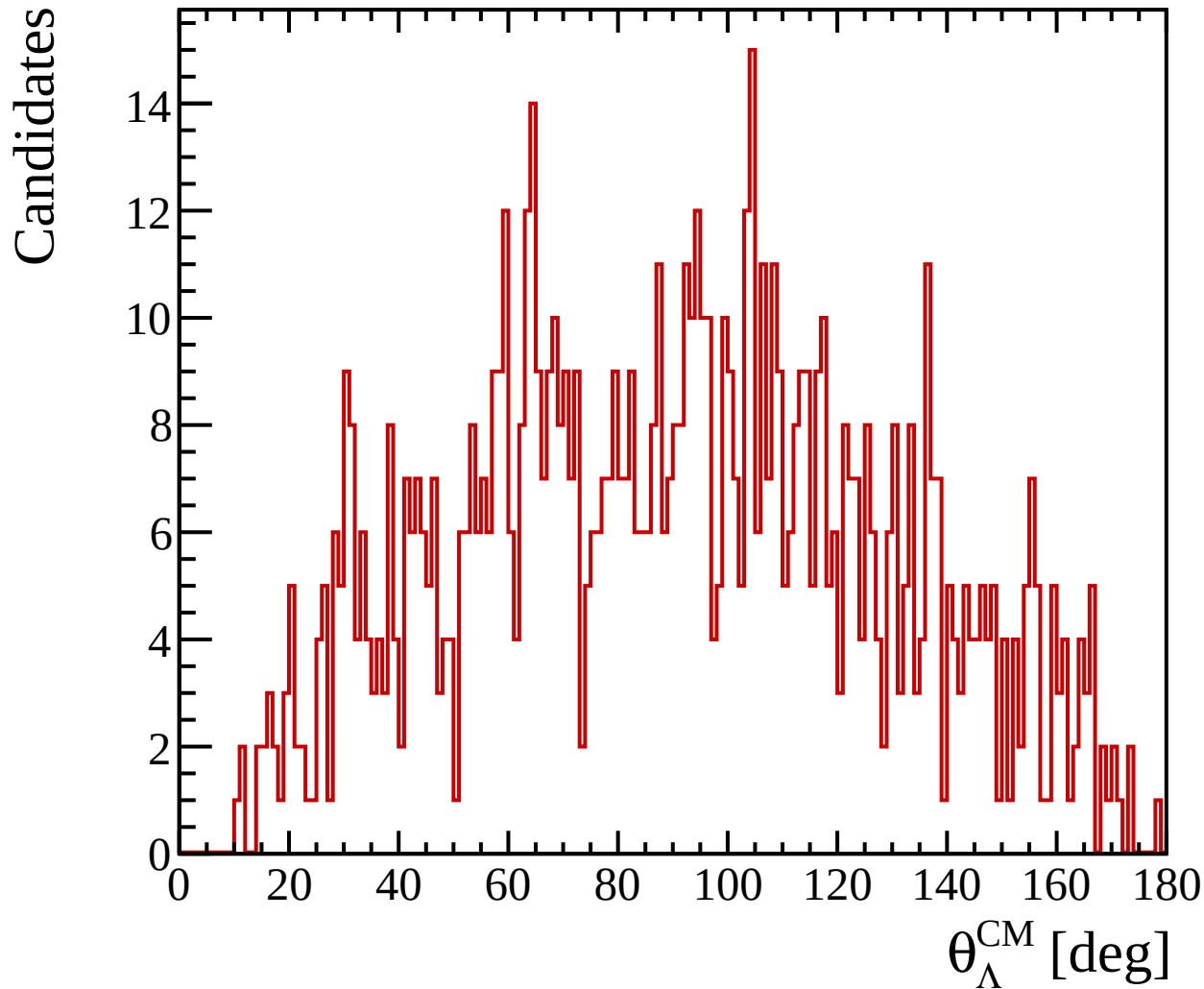
Λ production angle in K^-p CM



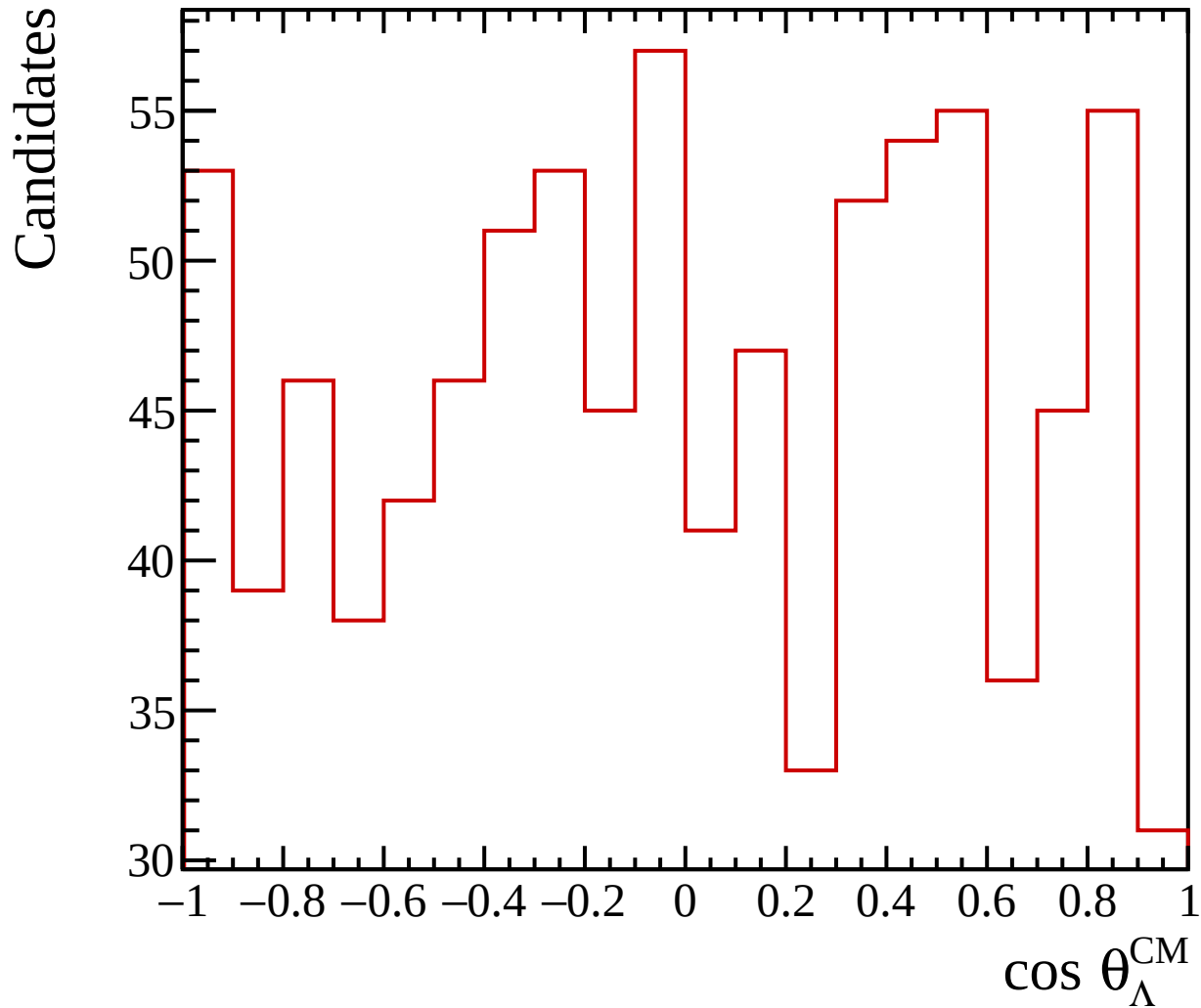
Selected Λ production angle ($M_X > 0.50$, LH2 inside)



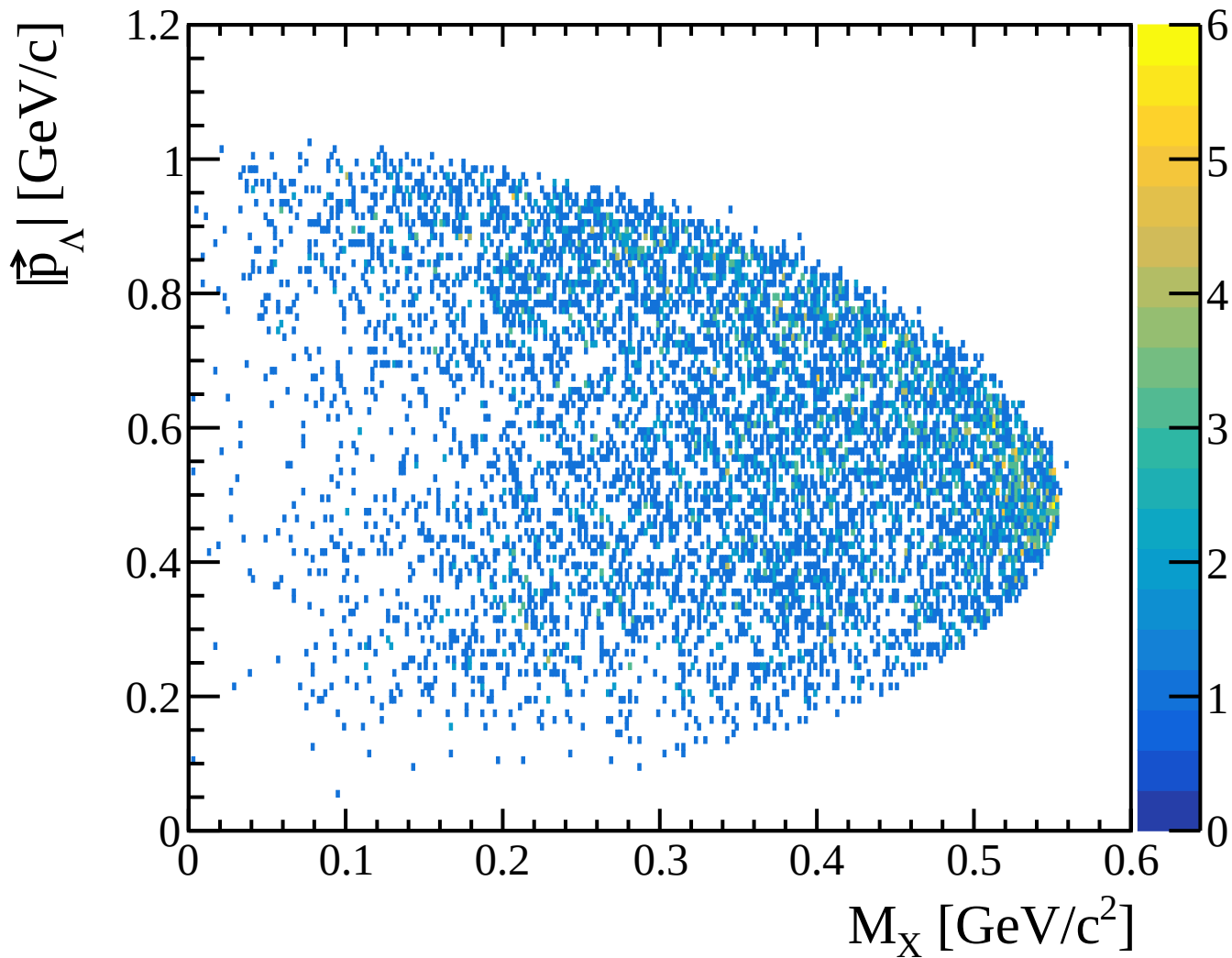
Selected Λ production angle ($M_X > 0.50$, LH2 inside)



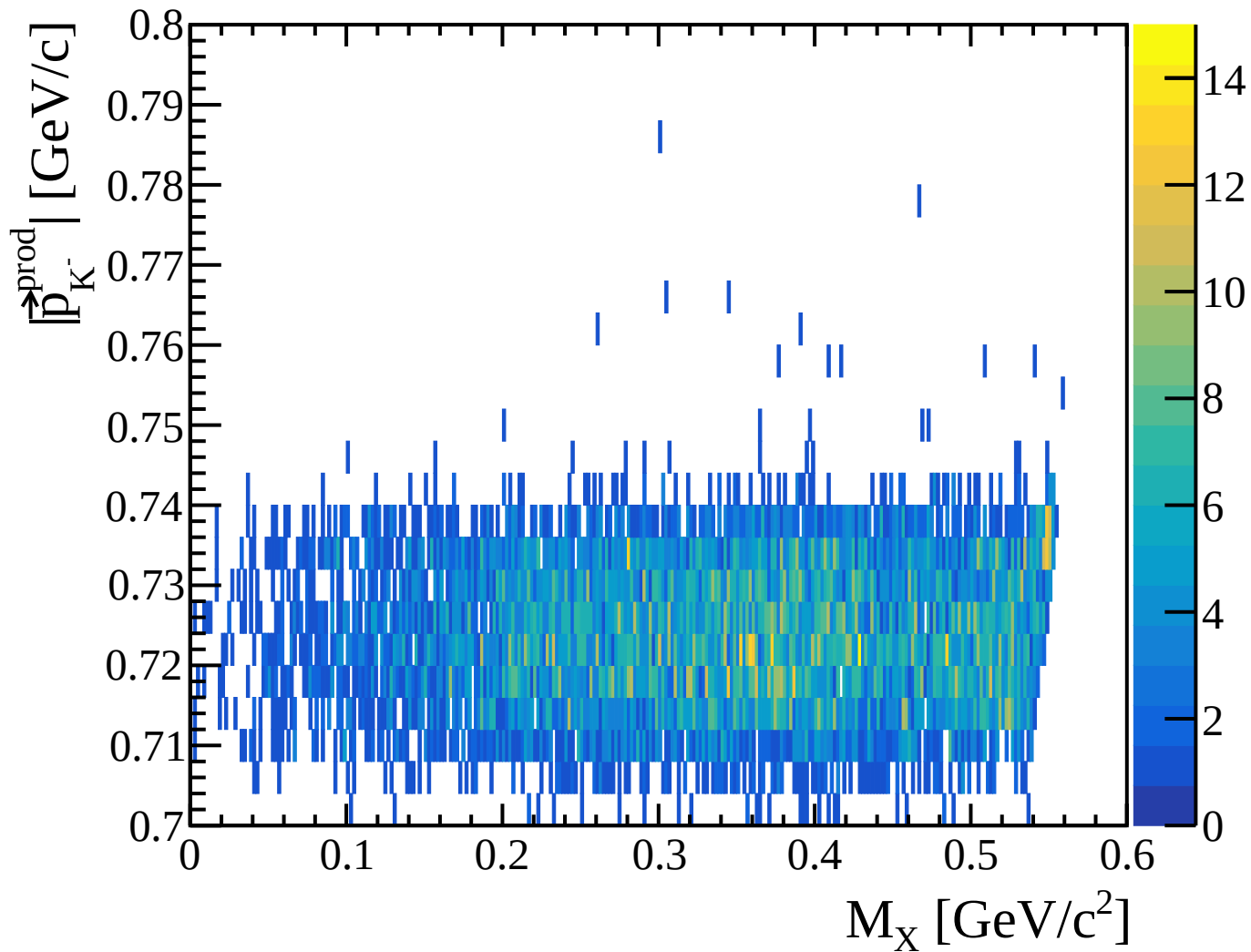
Selected Λ production angle ($M_X > 0.50$, LH2 inside)



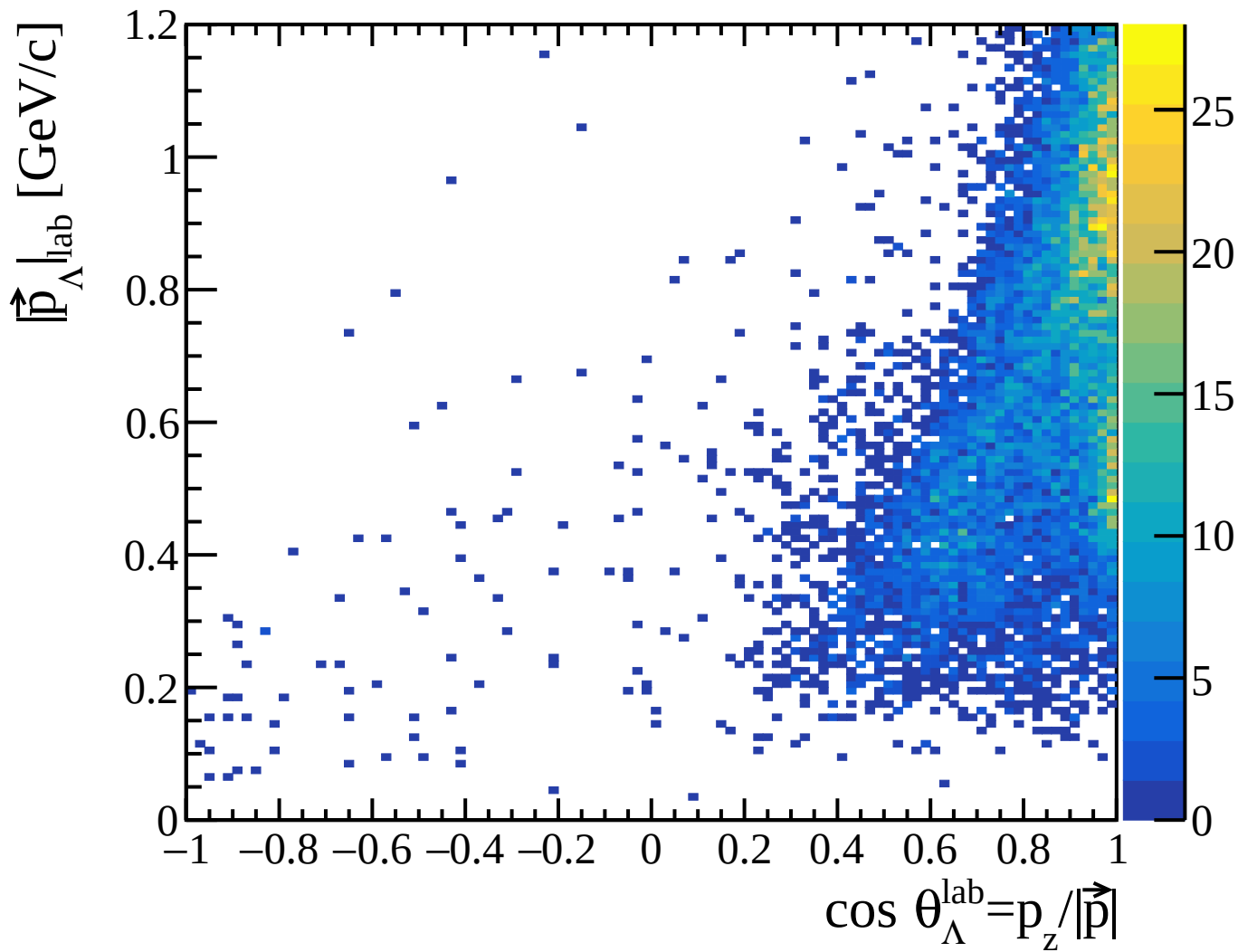
Missing mass vs reconstructed Λ momentum (LH2 inside, no M_X cut)



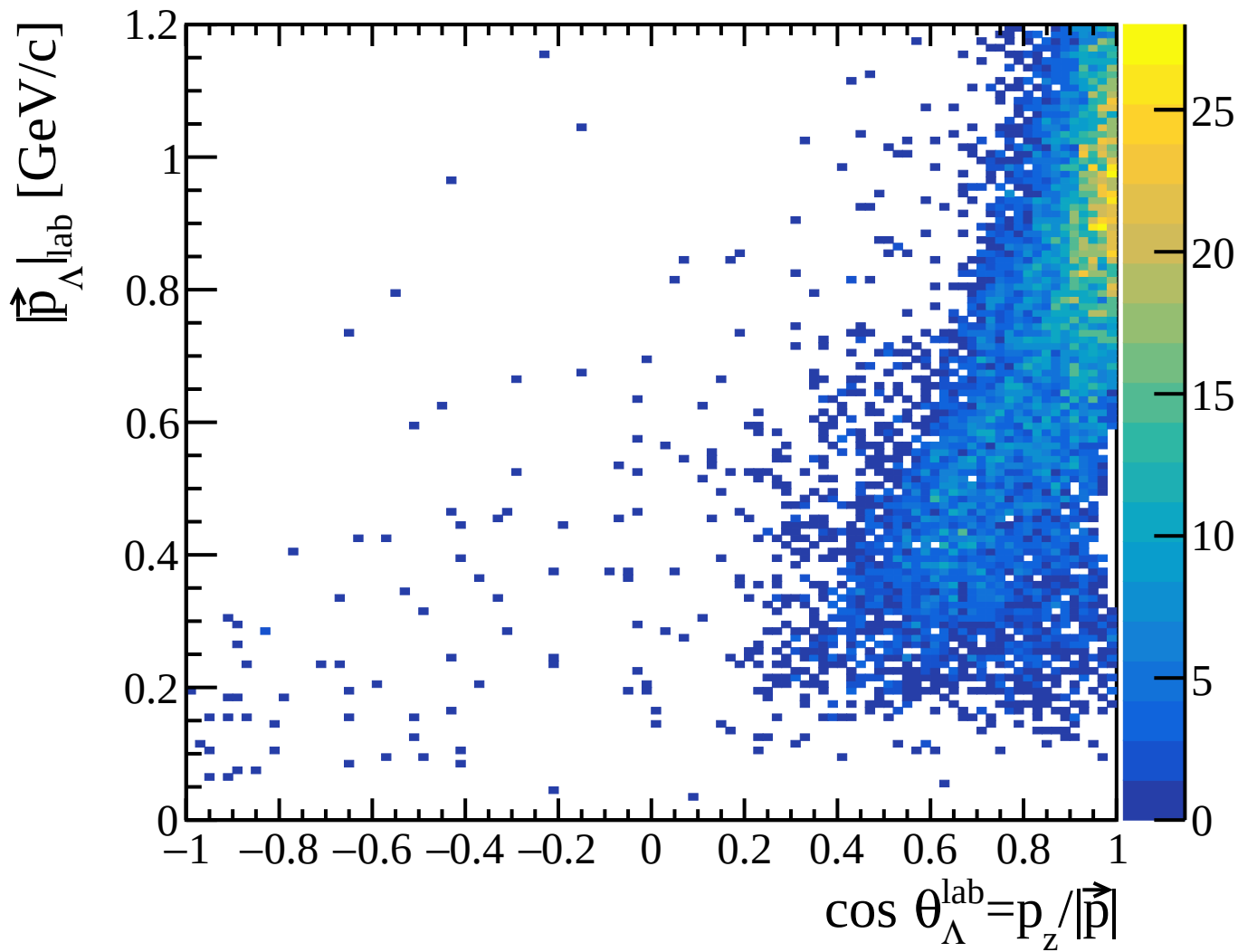
Missing mass vs production-vertex K18 momentum (LH2 inside, no M_x cut)



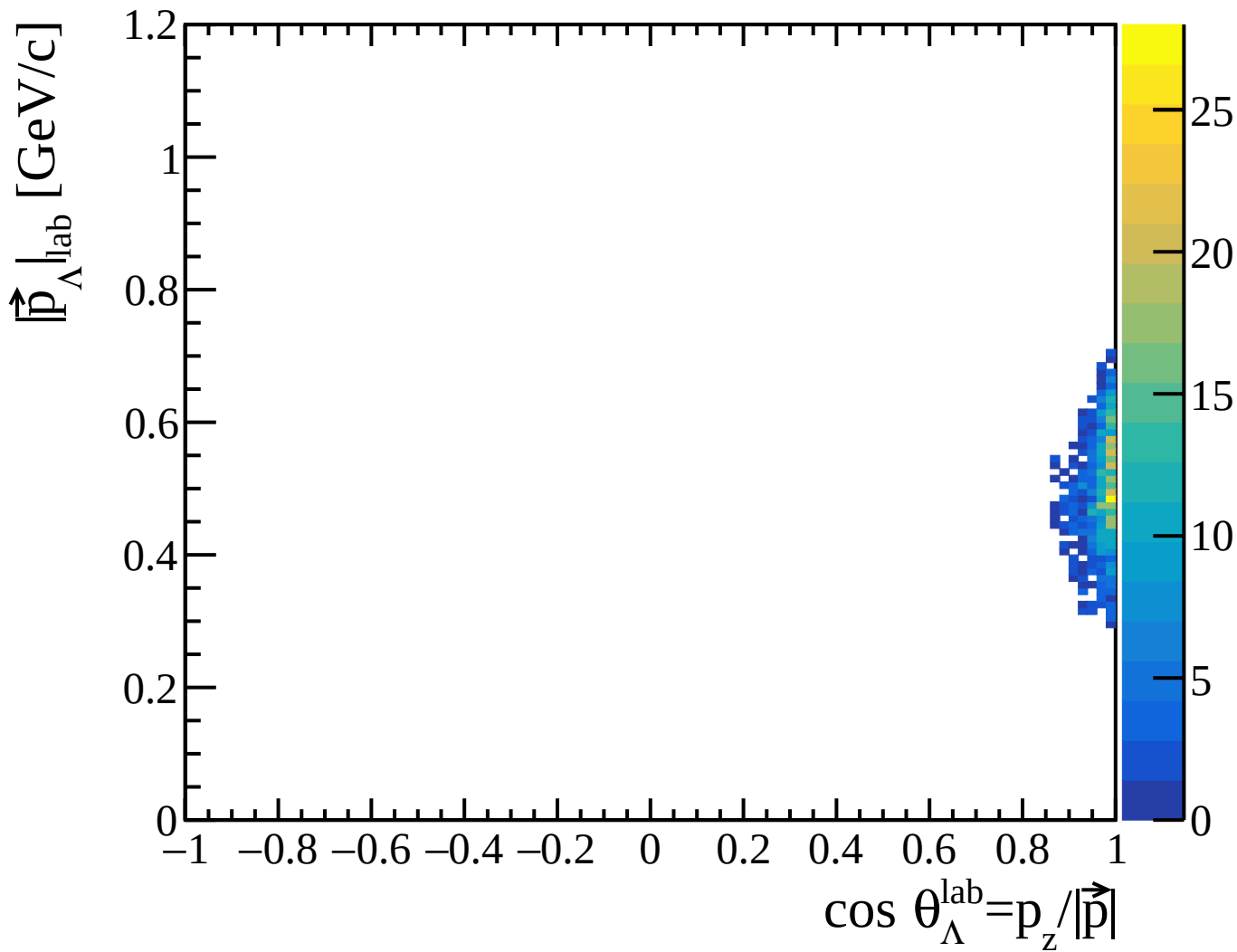
Selected Λ : lab $\cos\theta$ vs lab momentum (LH2 inside, all M_X)



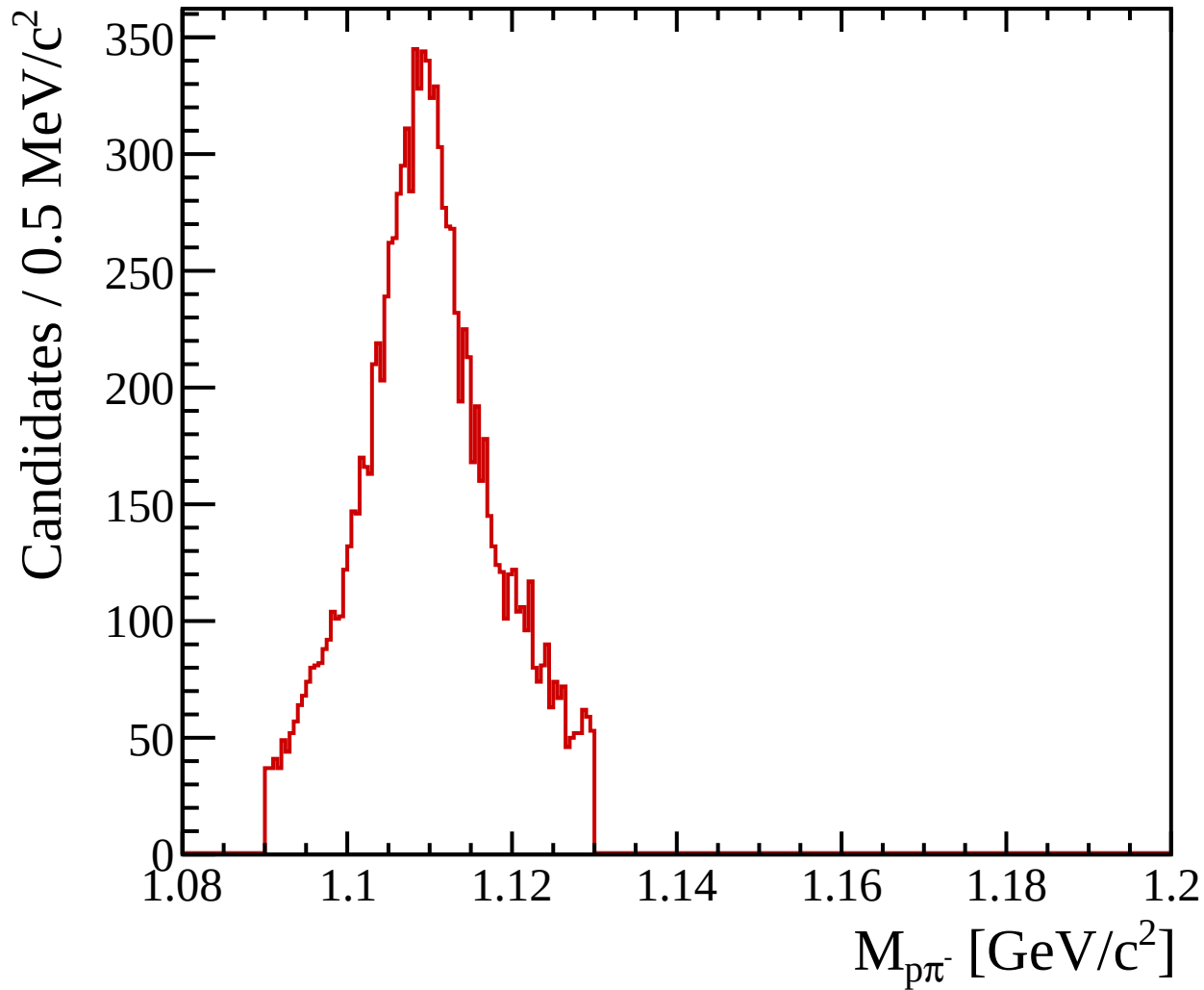
Selected Λ : lab $\cos\theta$ vs lab momentum (LH2 inside, $M_x \leq 0.50 \text{ GeV}/c^2$)



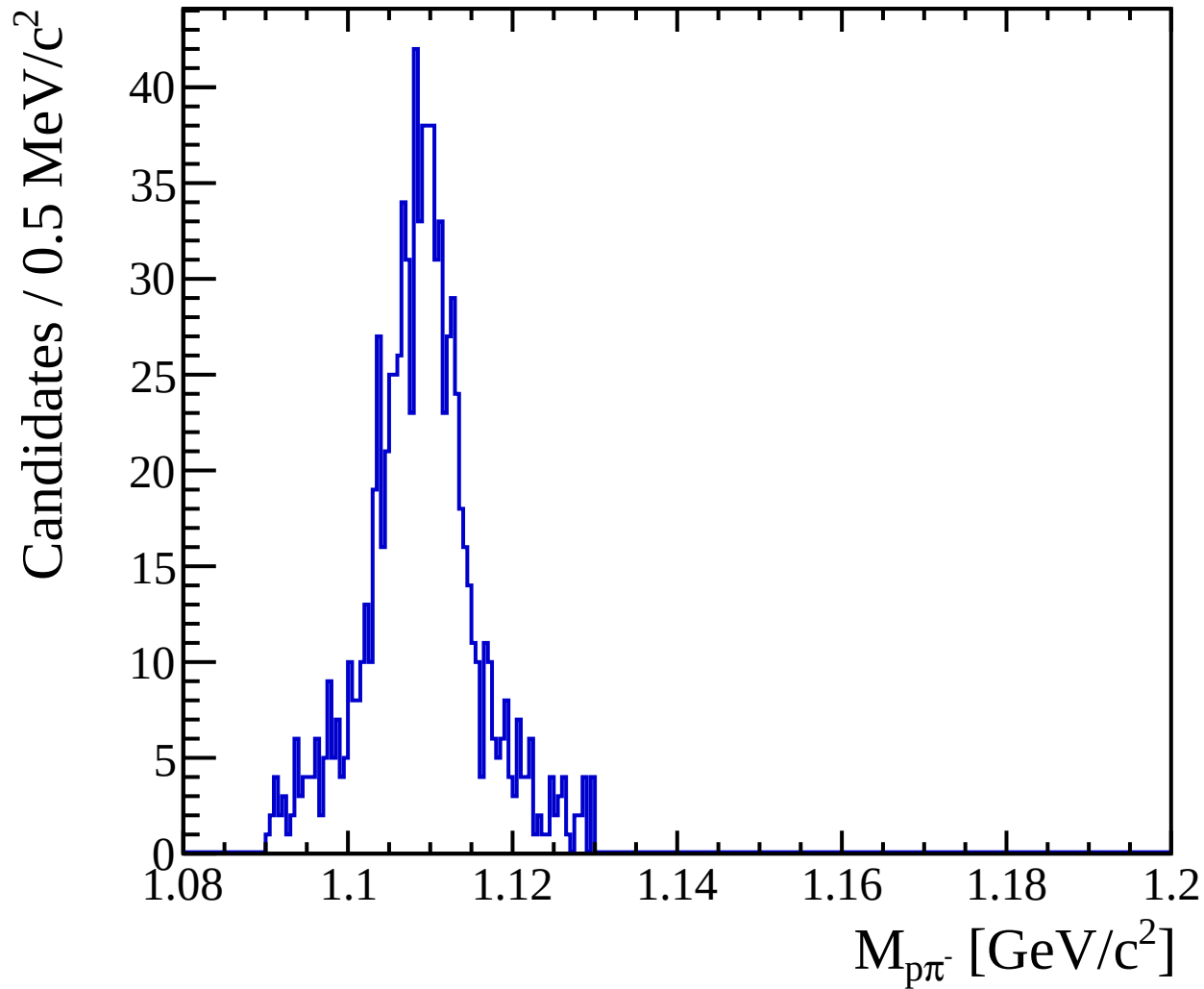
Selected Λ : lab $\cos\theta$ vs lab momentum (LH2 inside, $M_X > 0.50 \text{ GeV}/c^2$)

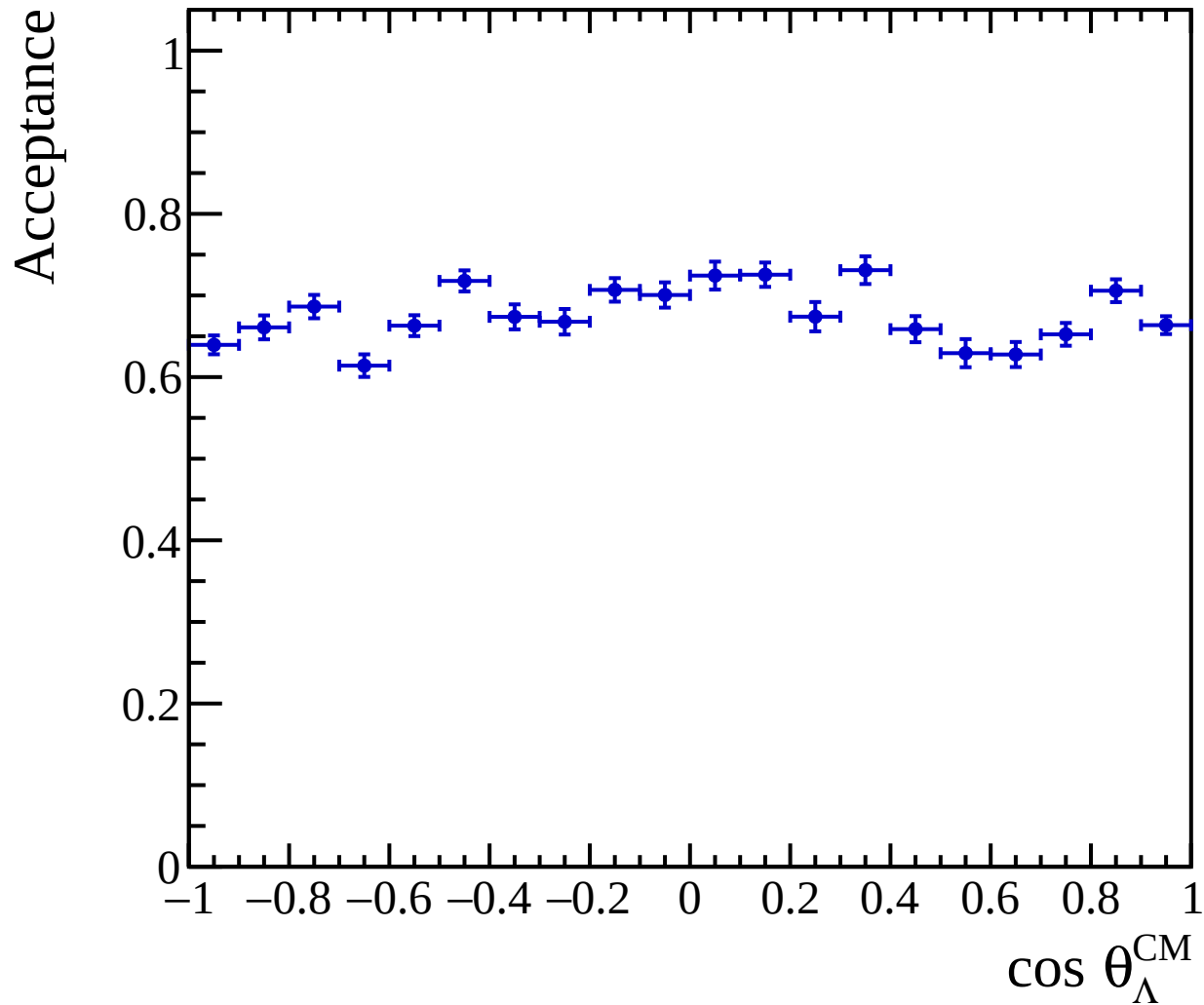


Selected Λ invariant mass (production vertex, LH2 inside)

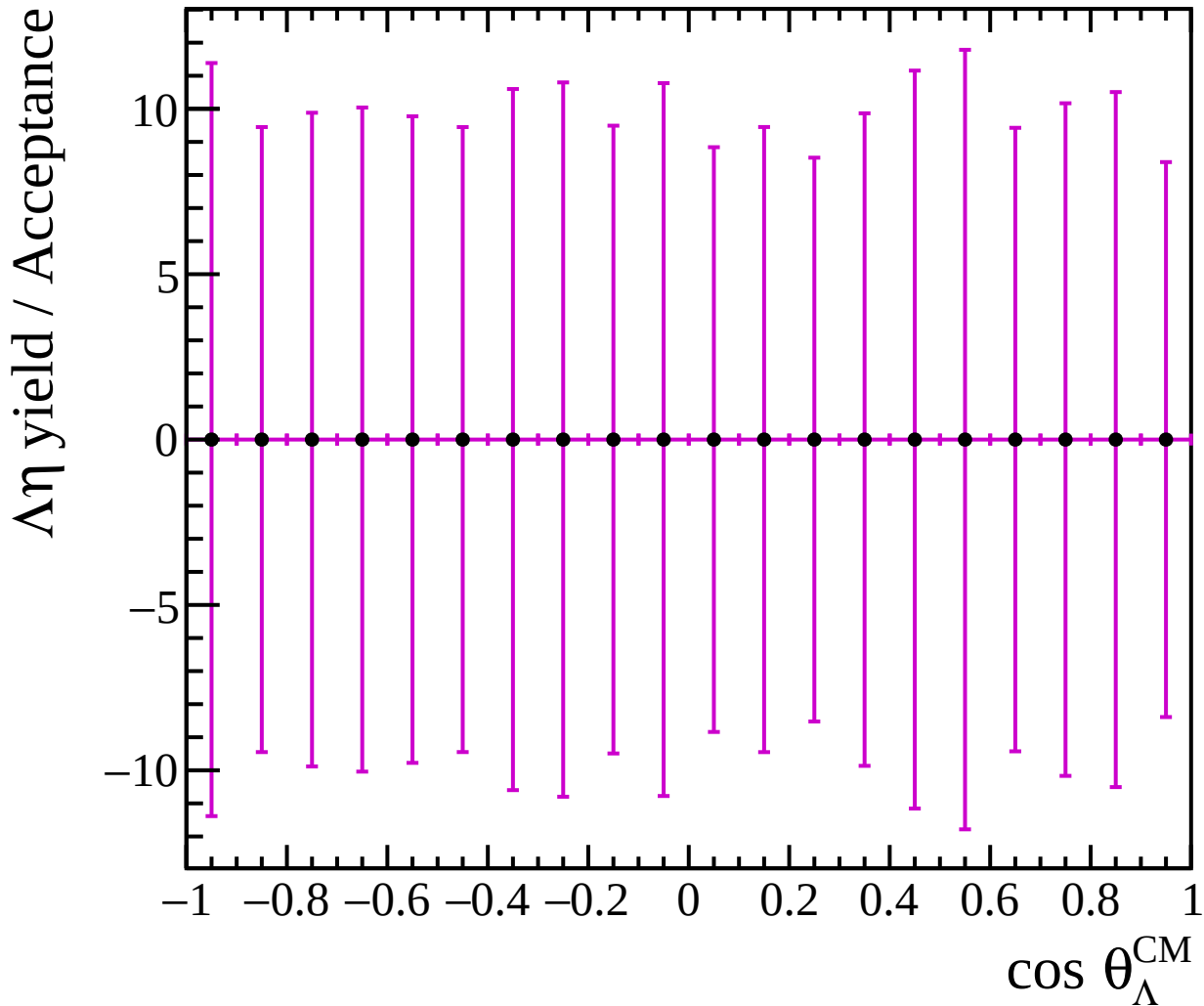


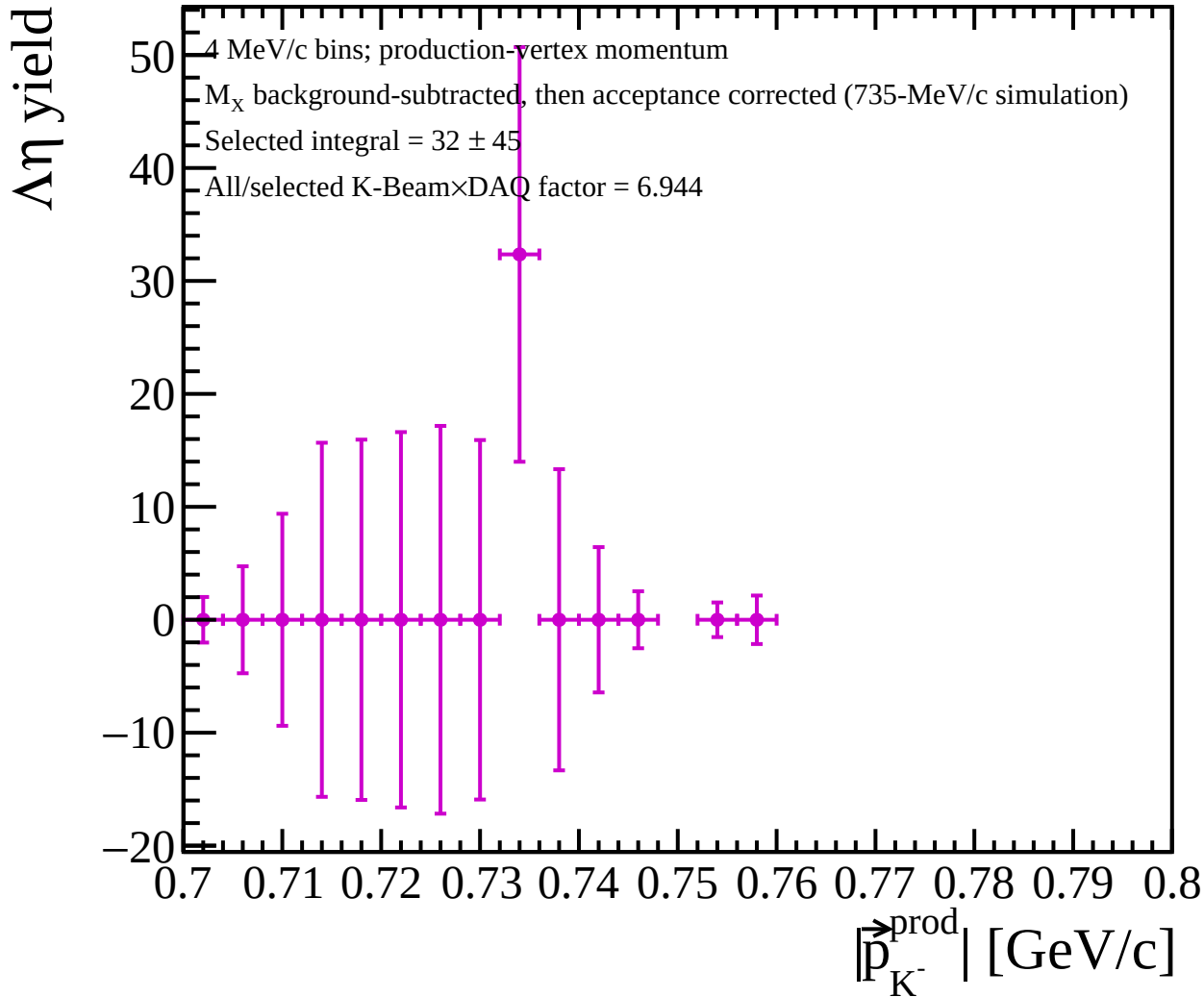
Selected Λ invariant mass (production vertex, LH2 inside, $M_x > 0.50$)



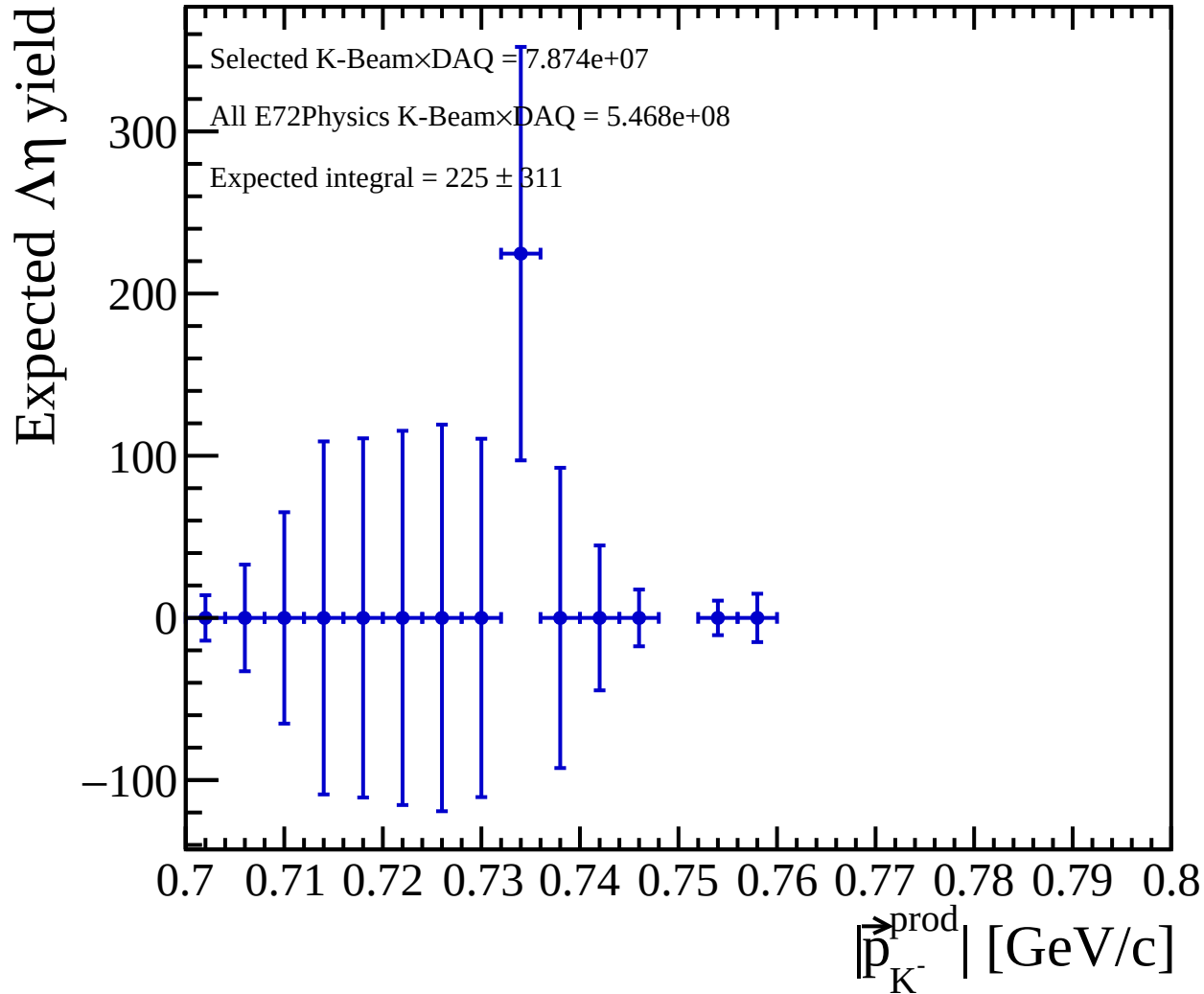
Trigger acceptance for generated $\Lambda\eta$ 

Background-subtracted $\Lambda\eta$ production angle, trigger-acceptance corrected (LH2 inside, $1.09 < M_{p\pi^-} < 1.13$)

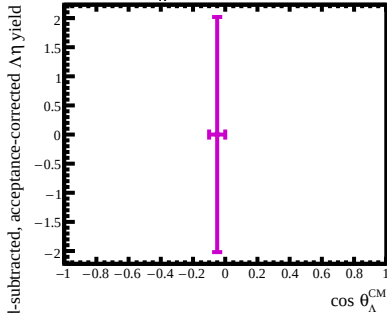




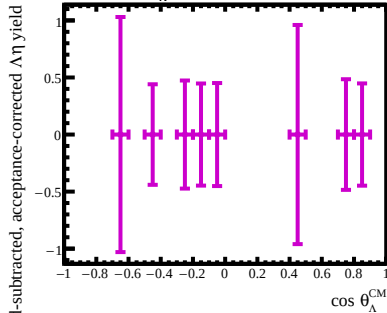
Expected $\Lambda\eta$ yield for all E72Physics runs (K-Beam scaled)



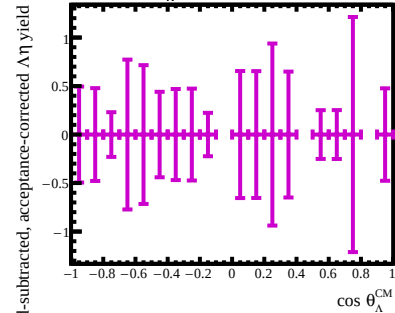
$p_K^{\text{prod}} = 700\text{--}704 \text{ MeV/c}$



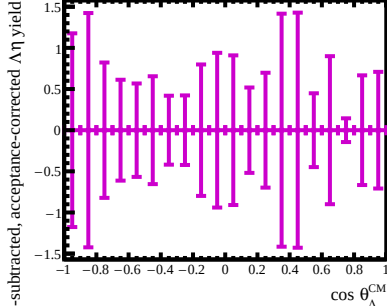
$p_K^{\text{prod}} = 704\text{--}708 \text{ MeV/c}$



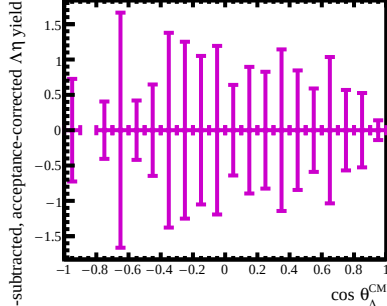
$p_K^{\text{prod}} = 708\text{--}712 \text{ MeV/c}$



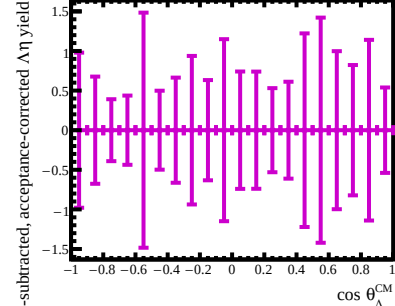
$p_K^{\text{prod}} = 712\text{--}716 \text{ MeV/c}$



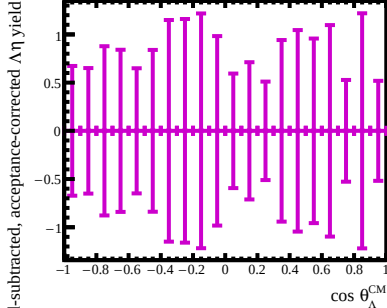
$p_K^{\text{prod}} = 716\text{--}720 \text{ MeV/c}$



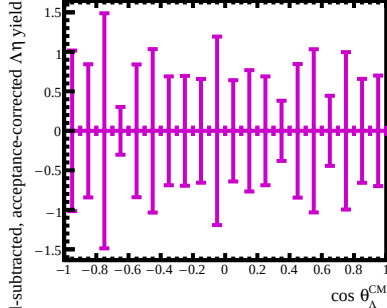
$p_K^{\text{prod}} = 720\text{--}724 \text{ MeV/c}$



$p_K^{\text{prod}} = 724\text{--}728 \text{ MeV/c}$



$p_K^{\text{prod}} = 728\text{--}732 \text{ MeV/c}$



$p_K^{\text{prod}} = 732\text{--}736 \text{ MeV/c}$

